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CHAPTER 1: INTRODUCTION

The proposed network architecture for the Alstom Networking Company is designed as a robust and scalable solution tailored to meet the diverse operational demands of a global enterprise. This comprehensive topology ensures seamless connectivity across geographically distributed branches while prioritizing fault tolerance, security, and high throughput to support critical business functions.

At the core of the infrastructure, the **Kuala Lumpur Headquarters** is engineered to handle substantial traffic loads. To satisfy high bandwidth and throughput requirements, EtherChannel technology is implemented to bundle physical links into logical aggregates. Furthermore, network resilience is secured through the Hot Standby Router Protocol (HSRP), which provides gateway redundancy and high availability to prevent service disruptions in the event of hardware failure.

The **Australia Branch** adopts a modern, mobility-first approach by implementing a complete Wireless LAN architecture managed by a centralized Wireless LAN Controller (WLC). To maintain strict access control, the branch utilizes WPA2-Enterprise security standards. This configuration mandates that all end devices undergo rigorous authentication against a dedicated RADIUS Server utilizing the IEEE 802.1X protocol before access is granted.

Centralized data services are consolidated at the **Remote India Branch**, where the primary Server Farm is strategically located within a Demilitarized Zone (DMZ) for enhanced security. This server farm hosts essential Web, DNS, FTP, and Email servers, acting as the central hub to facilitate efficient communication and secure file transfer operations throughout the entire Alstom organization.

Finally, the **Singapore Branch** is optimized to handle specialized, latency-sensitive traffic. A dedicated Voice VLAN is implemented to isolate telephony traffic, ensuring guaranteed call quality and network stability. Additionally, this branch features a local IoT Server designed to support smart security infrastructure, managing safety devices such as motion detectors and webcams.

CHAPTER 2: ENTIRE NETWORK LAYOUT

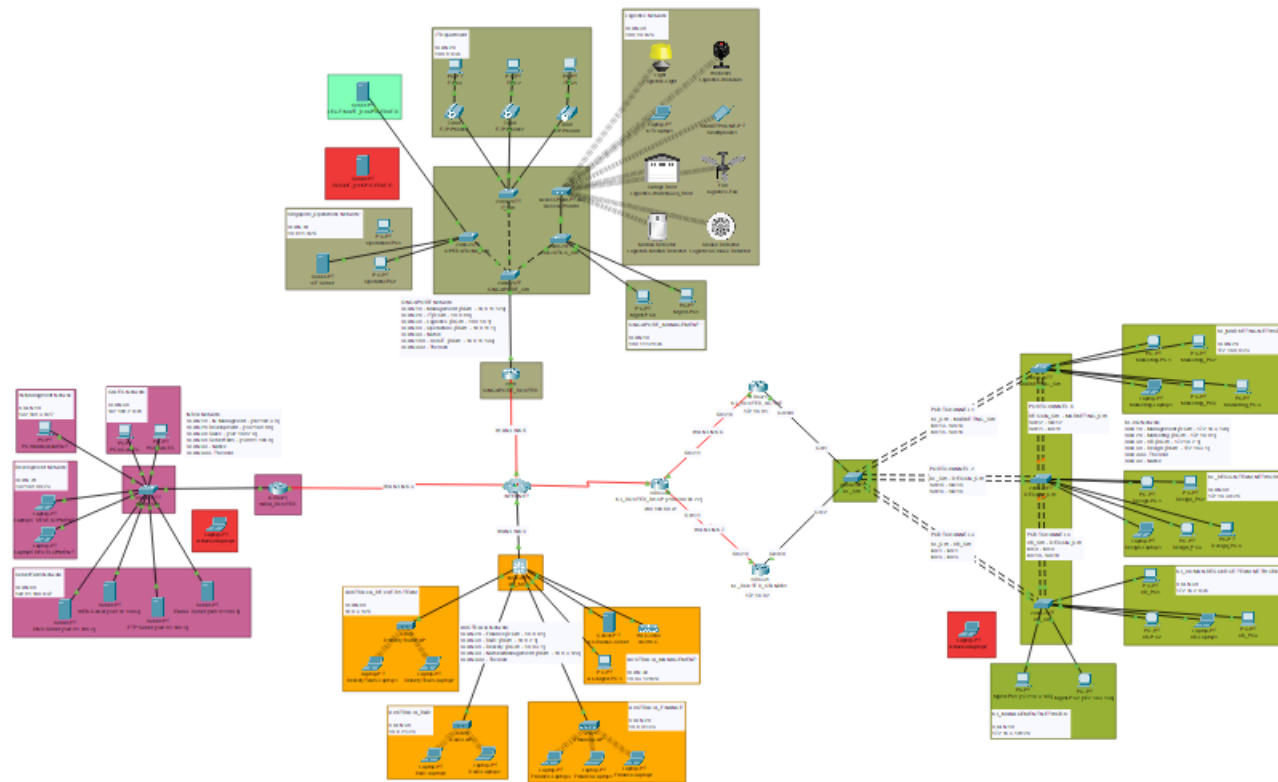


Figure 1 Overview of Alstom Network Topology

CHAPTER 3: IPv4 ADDRESSING TABLE

Area	Department	VLAN INFORMATION				Network ID	Subnet ID	Default Gateway	First Usable IP Address	Last Usable IP Address	Broadcast Address
		Network	No. of Hosts	VLAN ID	VLAN Name (Same as Network)						
	KUALA LUMPUR HQ										
Kuala Lumpur Branch	Marketing Team	Marketing	320	20	Marketing	172.16.0.0/23	255.255.254.0	172.16.0.1	172.16.0.1	172.16.1.254	172.16.1.255
	Human Resource Team	HR	200	30	HR	172.16.2.0/24	255.255.255.0	172.16.2.1	172.16.2.1	172.16.2.254	172.16.2.255
	Design Team	Design	110	40	Design	172.16.3.0/25	255.255.255.128	172.16.3.1	172.16.3.1	172.16.3.126	172.16.3.127
	Management	KL-Mgmt	55	10	KL-Mgmt	172.16.3.128/26	255.255.255.192	172.16.3.129	172.16.3.129	172.16.3.190	172.16.3.191
	Native	Native		99	Native						
	TheVoid	TheVoid		999	TheVoid						
	AUSTRALIA BRANCH										
Australia Branch (WLC)	Finance Team	Finance	410	20	Finance	10.0.0.0/23	255.255.254.0	10.0.0.1	10.0.0.1	10.0.1.254	10.0.1.255
	R & D Team	R&D	240	30	R&D	10.0.2.0/24	255.255.255.0	10.0.2.1	10.0.2.1	10.0.2.254	10.0.2.255
	Delivery Team	Delivery	100	40	Delivery	10.0.3.0/25	255.255.255.128	10.0.3.1	10.0.3.1	10.0.3.126	10.0.3.127
	Native&Management	Au-Mgmt&Native	40	99	Au-Mgmt&Native	10.0.3.128/26	255.255.255.192	10.0.3.129	10.0.3.129	10.0.3.190	10.0.3.191
	Junkyard	TheVoid		999	TheVoid						
	INDIA BRANCH										
India Branch	Development Team	Development	390	20	Development	192.168.0.0/23	255.255.254.0	192.168.0.1	192.168.0.1	192.168.1.254	192.168.1.255
	Sales Team	Sales	180	30	Sales	192.168.2.0/24	255.255.255.0	192.168.2.1	192.168.2.1	192.168.2.254	192.168.2.255
	Server Farm	ServerFarm	25	40	ServerFarm	198.81.100.0/27	255.255.255.224	198.81.100.1	198.81.100.1	198.81.100.30	198.81.110.31
	Management	IN-Mgmt	20	10	In-Mgmt	192.168.3.0/27	255.255.255.224	192.168.3.1	192.168.3.1	192.168.3.30	192.168.3.31
	Native	Native		99	Native						
	TheVoid	TheVoid		999	TheVoid						
	SINGAPORE BRANCH										
Singapore Branch	IT Services Team	IT	280	20	IT	10.0.8.0/23	255.255.254.0	10.0.8.1	10.0.8.1	10.0.9.254	10.0.9.255
	Logistics Team	Logistics	220	30	Logistics	10.0.10.0/24	255.255.255.0	10.0.10.1	10.0.10.1	10.0.10.255	10.0.11.0
	Operations Team	Operations	85	40	Operations	10.0.11.0/25	255.255.255.128	10.0.11.1	10.0.11.1	10.0.11.126	10.0.11.127
	Management	SI-Mgmt	45	10	SI-Mgmt	10.0.11.128/26	255.255.255.192	10.0.11.129	10.0.11.129	10.0.11.190	10.0.11.191
	VOICE	VOICE	25	150	VOICE	10.0.11.192/27	255.255.255.224	10.0.11.193	10.0.11.193	10.0.11.222	10.0.11.223
	Native	Native		99	Native						
	TheVoid	TheVoid		999	TheVoid						
	WAN LINKS										
WAN LINKS	-	WAN 1	2	-	-	200.100.50.0/30	255.255.255.252	-	200.100.50.1	200.100.50.2	200.100.50.3
	-	WAN 2	2	-	-	200.100.50.4/30	255.255.255.252	-	200.100.50.5	200.100.50.6	200.100.50.7
	-	WAN 3	2	-	-	200.100.50.8/30	255.255.255.252	-	200.100.50.9	200.100.50.10	200.100.50.11
	-	WAN 4	2	-	-	200.100.50.12/30	255.255.255.252	-	200.100.50.13	200.100.50.14	200.100.50.15
	-	WAN 5	2	-	-	200.100.50.16/30	255.255.255.252	-	200.100.50.17	200.100.50.18	200.100.50.19
HSRP	-	WAN 6	2	-	-	200.100.50.20/30	255.255.255.252	-	200.100.50.21	200.100.50.22	200.100.50.23
	-	WAN 7	2	-	-	200.100.50.24/30	255.255.255.252	-	200.100.50.25	200.100.50.26	200.100.50.27

Figure 2 Proposed IPv4 Addressing Table Designed for Alstom Networking Company

CHAPTER 4: LAN & WAN CONFIGURATION

4.1 KL Branch

4.1.1 Network Topology Overview

The KL branch is the main location (HQ) for Alstom. The project specification stipulates that this facility must be a fully wired network and thus ensuring the highest level of stability and security for the core business operations.

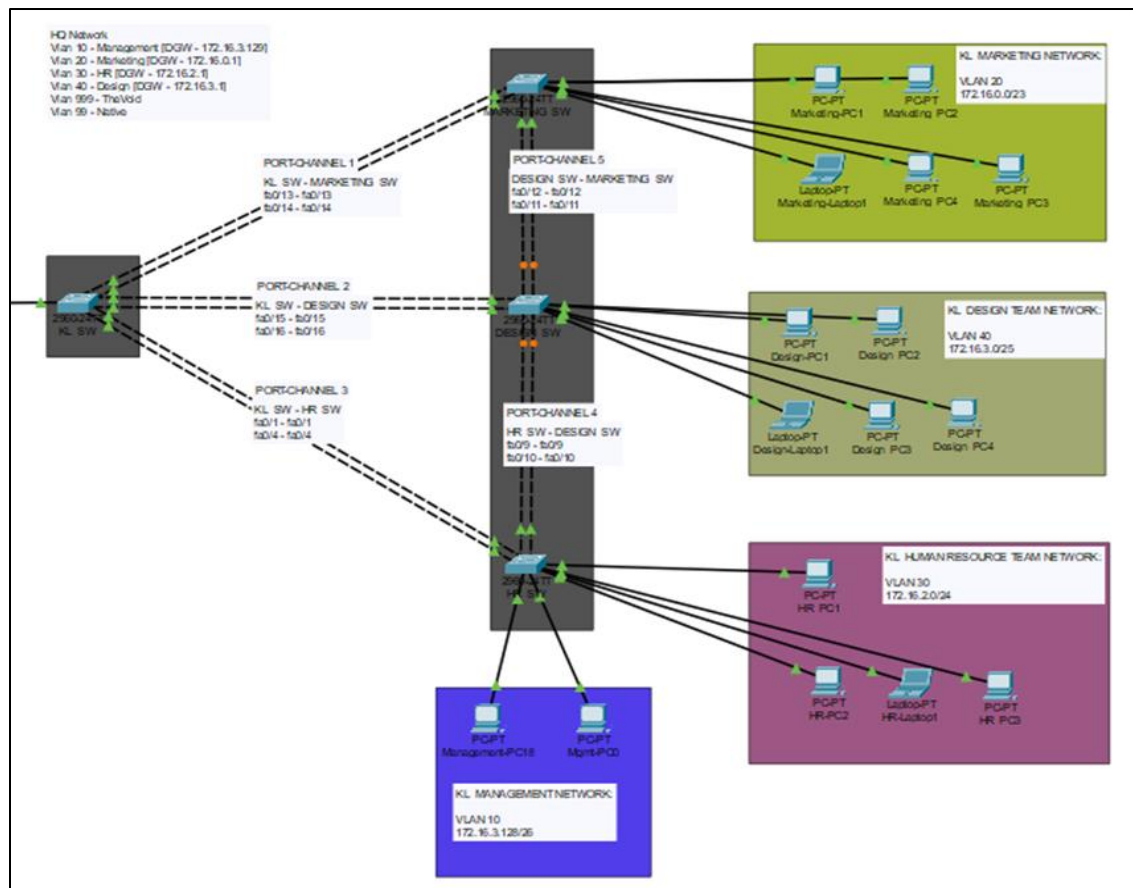


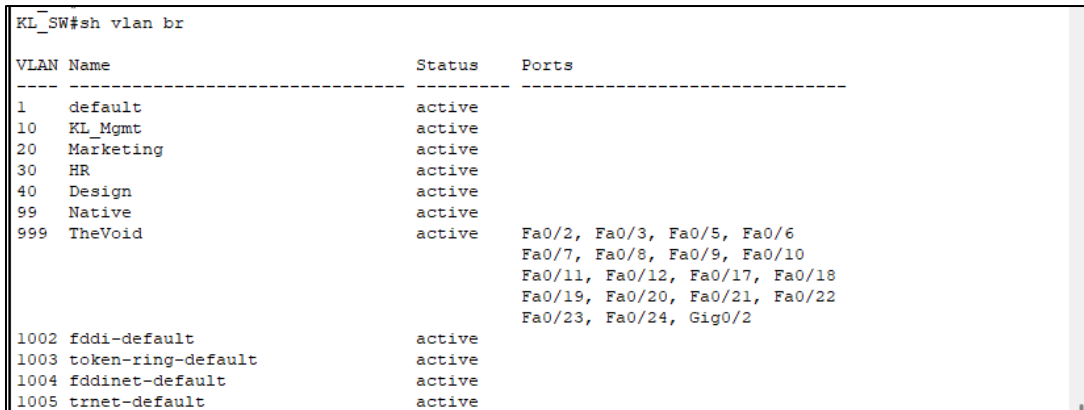
Figure 3 KL-HQ Network Topology Design

The KL topology adopts a hierarchical switching structure that revolves around the KL switch (Core/Distribution layer) interconnected with three access switches that are designated for different departments: MARKETING SW, DESIGN SW, and HR SW.

Redundancy & Performance: To ensure high availability and increased bandwidth between switches, EtherChannel (see Figure 1, port-channels 1, 2, and 3) has been implemented between the KL SW and the departmental switches.

Inter-VLAN Connectivity: Physical links between the departmental switches (see Figure 1, port-channels 4 and 5) suggest a redundant mesh or ring-style topology to mitigate single points of failure, supporting the HSRP requirements for gateway redundancy.

4.1.2 VLAN Implementation



VLAN	Name	Status	Ports
1	default	active	
10	KL_Mgmt	active	
20	Marketing	active	
30	HR	active	
40	Design	active	
99	Native	active	
999	TheVoid	active	Fa0/2, Fa0/3, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Figure 4 KL-HQ VLAN Configuration

Departmental Segmentation: The “show vlan brief” verification provides the confirmation that the KL headquarters has been reasonably separated into four essential departments: VLAN 10 (KL_Mgmt), VLAN 20 (Marketing), VLAN 30 (HR), and VLAN 40 (Design) (see Figure 2).

VLAN 999 (TheVoid): This is a "Black Hole" VLAN. All the ports that are not being used (like Fa0/2, Fa0/3, and Fa0/5-12) get the VLAN number assigned and shut down manually. This is an important preventive measure against the mentioned types of attacks, as it makes the process of gaining access to the network through an open wall jack very difficult for the attackers.

Active Status: All the VLANs for the departments are marked as "active," which is a clear indication that the traffic for these segments is properly managed by the database configuration.

4.1.3 DHCP Pool Deployment

```

KL_ROUTER#sh ip dhcp pool

Pool KL_Mgmt-vlan10 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next)    : 0 / 0
  Total addresses              : 62
  Leased addresses            : 2
  Excluded addresses          : 4
  Pending event               : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  172.16.3.129      172.16.3.129 - 172.16.3.190  2 / 4 / 62

Pool Marketing-vlan20 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next)    : 0 / 0
  Total addresses              : 510
  Leased addresses            : 5
  Excluded addresses          : 4
  Pending event               : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  172.16.0.1         172.16.0.1 - 172.16.1.254  5 / 4 / 510

Pool HR-vlan30 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next)    : 0 / 0
  Total addresses              : 254
  Leased addresses            : 4
  Excluded addresses          : 4
  Pending event               : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  172.16.2.1         172.16.2.1 - 172.16.2.254  4 / 4 / 254

Pool Design-vlan40 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next)    : 0 / 0
  Total addresses              : 126
  Leased addresses            : 5
  Excluded addresses          : 4
  Pending event               : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  172.16.3.1         172.16.3.1 - 172.16.3.126  5 / 4 / 126
KL_ROUTER#
KL_ROUTER#
KL_ROUTER#

```

Figure 5 KL-HQ IP DHCP Pool Configuration

In order to make the network at the Alstom HQ and the distant branches more efficient, DHCPv4 was introduced to automatically distribute IP addresses. By doing so, it removes the need for manual configuration which is more time-consuming and prone to mistakes, such as assigning the same IP to two devices which can cause the network to go off for a while. As per the scenario needs, centralized DHCP services were established on the core layer devices to support several VLANs in different departments such as KL_Management, KL_HR, and KL_Design.

4.1.4 Interface Trunking and VLAN Tagging

```

KL_SW#sh int tr
Port      Mode      Encapsulation  Status      Native vlan
Po1       on         802.1q         trunking    99
Po2       on         802.1q         trunking    99
Po3       on         802.1q         trunking    99
Gig0/1    on         802.1q         trunking    99

Port      Vlans allowed on trunk
Po1       10,20,30,40,99
Po2       10,20,30,40,99
Po3       10,20,30,40,99
Gig0/1    10,20,30,40,99

Port      Vlans allowed and active in management domain
Po1       10,20,30,40,99
Po2       10,20,30,40,99
Po3       10,20,30,40,99
Gig0/1    10,20,30,40,99

Port      Vlans in spanning tree forwarding state and not pruned
Po1       10,20,30,40,99
Po2       10,20,30,40,99
Po3       10,20,30,40,99
Gig0/1    10,20,30,40,99

```

Figure 6 KL-HQ Switch Interface Trunk Configuration

At the Alstom HQ network, the IEEE 802.1Q trunking has been deployed on the KL_SW for the smooth and unstoppable communication of departmental units and centralized resources like the India-based Server Farm. The use of logical EtherChannel interfaces (see Figure 6, Po1, Po2, and Po3) that carry multiple VLAN IDs (see Figure 6, VLANs 10, 20, 30, 40, and 99) over the core links is indicated by the show interface trunk output. This arrangement permits the switches to share the data belonging to various departments through a single physical or logical link without needless frame duplication, and each frame is properly tagged with its associated VLAN ID to uphold logical separation. Moreover, the arrangement not only designates VLAN 99 as the Native VLAN but also counters possible security threats like VLAN hopping, thus ensuring the KL_ROUTER's role in inter-VLAN routing that all cross-departmental and server-bound traffic gets inspected and directed through the default gateways marked for that purpose.

4.1.5 OSPF Dynamic Routing Protocol

```

KL_ROUTER#sh ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    172.16.0.0 0.0.1.255 area 0
    172.16.2.0 0.0.0.255 area 0
    172.16.3.0 0.0.0.127 area 0
    172.16.3.128 0.0.0.63 area 0
    200.100.50.0 0.0.0.3 area 0
  Passive Interface(s):
    Vlan1
    GigabitEthernet0/0/0
    GigabitEthernet0/0/1
    Serial0/1/1
    Serial0/2/0
    Serial0/2/1
    GigabitEthernet0/0/0.10
    GigabitEthernet0/0/0.20
    GigabitEthernet0/0/0.30
    GigabitEthernet0/0/0.40
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1           110          00:02:37
    2.2.2.2           110          00:02:36
    3.3.3.3           110          00:02:39
    4.4.4.4           110          00:02:36
    5.5.5.5           110          00:02:46
  Distance: (default is 110)

```

Figure 7 KL-HQ OSPF Routing Configuration

The OSPF (Open Shortest Path First) dynamic routing protocol is utilized by the network for seamless interconnection between the KL Headquarters and remote branches in Australia, Singapore, and India. The output of “show ip protocol” gives the indication that the KL_ROUTER has been assigned the Router ID of 1.1.1.1 and is participating in Area 0, where also the advertised subnets of Management, HR, and Design get merged into the global routing table. In accordance with the assignment's requirement of public IP addressing on WAN links, the serial interface connecting to the provider cloud uses the network of 200.100.50.0/30. Moreover, a very important security strategy was put in place by having all LAN-facing sub-interfaces set up as passive interfaces. This approach not only fortifies the security of the organization but also eliminates the possibilities that OSPF hello packets could be sent into user segments thereby effectively reducing Routing Protocol Spoofing and reconnaissance attacks and also saving router CPU and bandwidth resources.

4.2 India Branch

4.2.1 Network Topology Overview

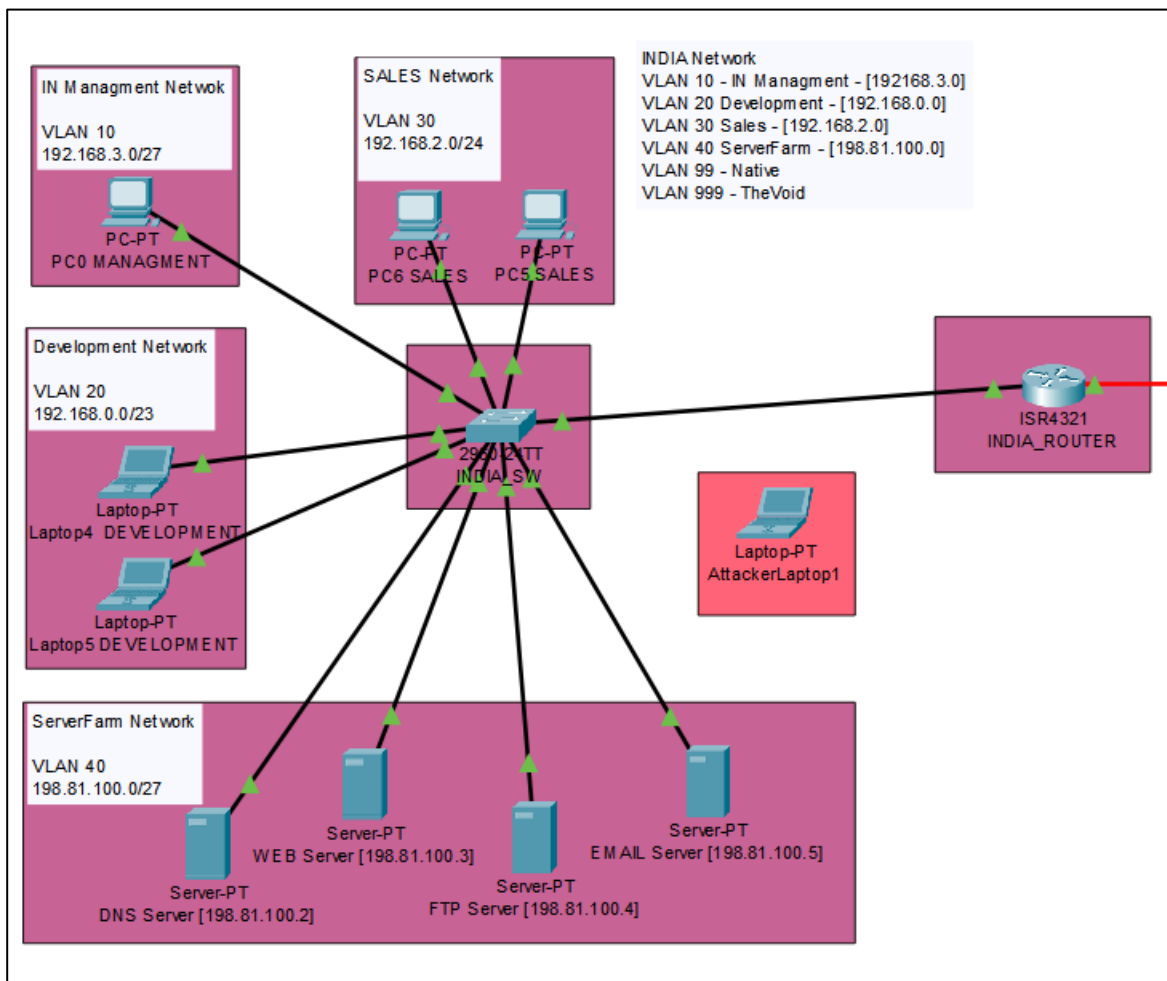


Figure 8 Overview of India Network Topology

The INDIA branch is a remote location of operation of Alstom, and has been designed as a fully wired network, and in strict observance of the project requirements. It was also strategically chosen to use a wired-only implementation to ensure the connectivity was stable, the performance was predictable, and that security was improved in its critical departments and the server farm at this location.

The Router-on-a-Stick design has a cost effective and scalable solution to fit a remote branch setting. The design allows the hardware to be simplified in nature yet dynamic in IP allocation, centralized routing, and safe access to internal services like DNS, FTP, Email, and web servers housed within the server farm.

4.2.2 VLAN Implementation

```
INDIA_SW#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	IN-Mgmt	active	Fa0/9
20	Development	active	Fa0/1, Fa0/2
30	Sales	active	Fa0/3, Fa0/4
40	ServerFarm	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8
99	Native	active	
999	TheVoid	active	Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
INDIA_SW#
```

Figure 9 VLAN configuration of India Switch

In the India branch The default VLAN 1 is currently active but notably contains no assigned ports. This is a significant security measure, as removing ports from the default VLAN helps prevent unauthorized access and potential VLAN hopping attacks.

Security is further reinforced by the handling of unused interfaces. All spare ports (Fa0/10 through Fa0/24 and Gig0/2) have been moved into a dedicated isolation VLAN named "TheVoid" (VLAN 999). As per security best practices, these ports are also administratively shut down to ensure they cannot be used if someone physically plugs into them.

Active network traffic is segmented into specific functional departments to organize data flow. This configuration includes VLAN 10 (IN-Mgmt.), VLAN 20 (Development), and VLAN 30 (Sales), which host the active user and management connections.

The bottom of the list displays VLANs 1002 through 1005 (fddi-default, token-ring-default, etc.). These are immutable legacy VLANs that are automatically created by the Cisco IOS for backward compatibility with older technologies and are not used in this modern deployment.

4.2.3 DHCP Pool Deployment

The Show ip dhcp pool command is issued on the India router in the figure below to ensure there is an implementation of the DHCPv4 configuration to the India branch network. The output obtained below confirms that there are several DHCP pools that have successfully created to facilitate the process of automatic allocation of IP addresses across the different VLANs that are in use.

```
INDIA_ROUTER#sh ip dhcp pool

Pool In-Mgmt :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 30
Leased addresses                  : 1
Excluded addresses                : 8
Pending event                    : none

1 subnet is currently in the pool
Current index   IP address range      Leased/Excluded/Total
192.168.3.1     192.168.3.1 - 192.168.3.30 1 / 8 / 30

Pool Sales :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 254
Leased addresses                  : 2
Excluded addresses                : 8
Pending event                    : none

1 subnet is currently in the pool
Current index   IP address range      Leased/Excluded/Total
192.168.2.1     192.168.2.1 - 192.168.2.254 2 / 8 / 254

Pool Development :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 510
Leased addresses                  : 2
Excluded addresses                : 8
Pending event                    : none

1 subnet is currently in the pool
Current index   IP address range      Leased/Excluded/Total
192.168.0.1     192.168.0.1 - 192.168.1.254 2 / 8 / 510
```

Figure 10 India DHCP Pool Configuration

The DHCPv4 pool check allows verifying that the centralized DHCP services on the India router is properly operating. Such implementation eases the process to manage the IP addresses, minimize the configuration errors and maintain the reliable and uniform network connectivity throughout the entire VLANs in the India branch.

4.2.4 Interface Trunking and VLAN Tagging

```

INDIA_SW#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    99

Port      Vlans allowed on trunk
Gig0/1    10,20,30,40,99

Port      Vlans allowed and active in management domain
Gig0/1    10,20,30,40,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    10,20,30,40,99

INDIA_SW#

```

Figure 11 India Switch Interface Trunk Information

The show interface trunk output for the INDIA_SW switch confirms that port GigabitEthernet0/1 is actively operating as a trunk link. The mode is listed as "on," indicating a static configuration that forces the port into trunking state without relying on dynamic negotiation. Furthermore, the interface utilizes the 802.1q encapsulation protocol to tag frames, ensuring compatibility with standard VLAN implementations.

The configuration explicitly defines VLAN 99 as the Native VLAN for this trunk interface. By designating a specific VLAN for untagged traffic, the switch separates administrative or control traffic from standard user data. This is a common security practice intended to mitigate VLAN hopping attacks and ensure that untagged frames are not inadvertently processed on the default VLAN.

The trunk is configured with a strict allow list, permitting only VLANs 10, 20, 30, 40, and 99 to traverse the link. Significantly, the Default VLAN 1 and the Blackhole VLAN 999 are absent from the allowed list. This exclusion demonstrates that traffic from these VLANs is blocked, effectively reducing the broadcast domain, and preventing unauthorized access via the default or unused VLANs.

4.2.5 OSPF Dynamic Routing Protocol

```
INDIA_ROUTER>sh ip protocol

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 4.4.4.4
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.0.0 0.0.1.255 area 0
    192.168.2.0 0.0.0.255 area 0
    192.168.3.0 0.0.0.31 area 0
    198.81.100.0 0.0.0.31 area 0
    200.100.50.8 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:01:15
    2.2.2.2          110          00:01:15
    3.3.3.3          110          00:01:10
    4.4.4.4          110          00:01:16
    5.5.5.5          110          00:09:13
  Distance: (default is 110)

INDIA_ROUTER>
```

Figure 12 India Router OSPF Routing Protocol Information

The show ip protocol verification command shows that the router in India is actively running OSPF process ID10 and it is part of a single OSPF area, i.e., Area0. The router has been assigned with the Router ID value of 4.4.4.4 that is used to identify it uniquely in the OSPF domain as well as enabling it to establish the correct neighbor relationship with other routers over the entire enterprise network.

The verification output shows that there is the advertisement of multiple internal network subnets into Area 0; namely 192.168.0.0/24, 192.168.2.0/24, 192.168.3.0/27, and 198.81.100.0/27. This proves that all the VLANs and the server-farm network in the branch at India are all well-integrated into the OSPF routing domain. In addition, the public wide-area net 200.100.50.8/30 is also announced and thus, provides end-to-end connectivity between the India branch and other remote locations.

Comprehensively, the OSPF protocol validation reveals India branch router is fully set up to work with dynamic routing hence facilitating scale, resiliency, and safety of inter-branch communication as per laid down best practices of enterprise-networking.

4.3 Australia Branch

4.3.1 Network Topology Overview

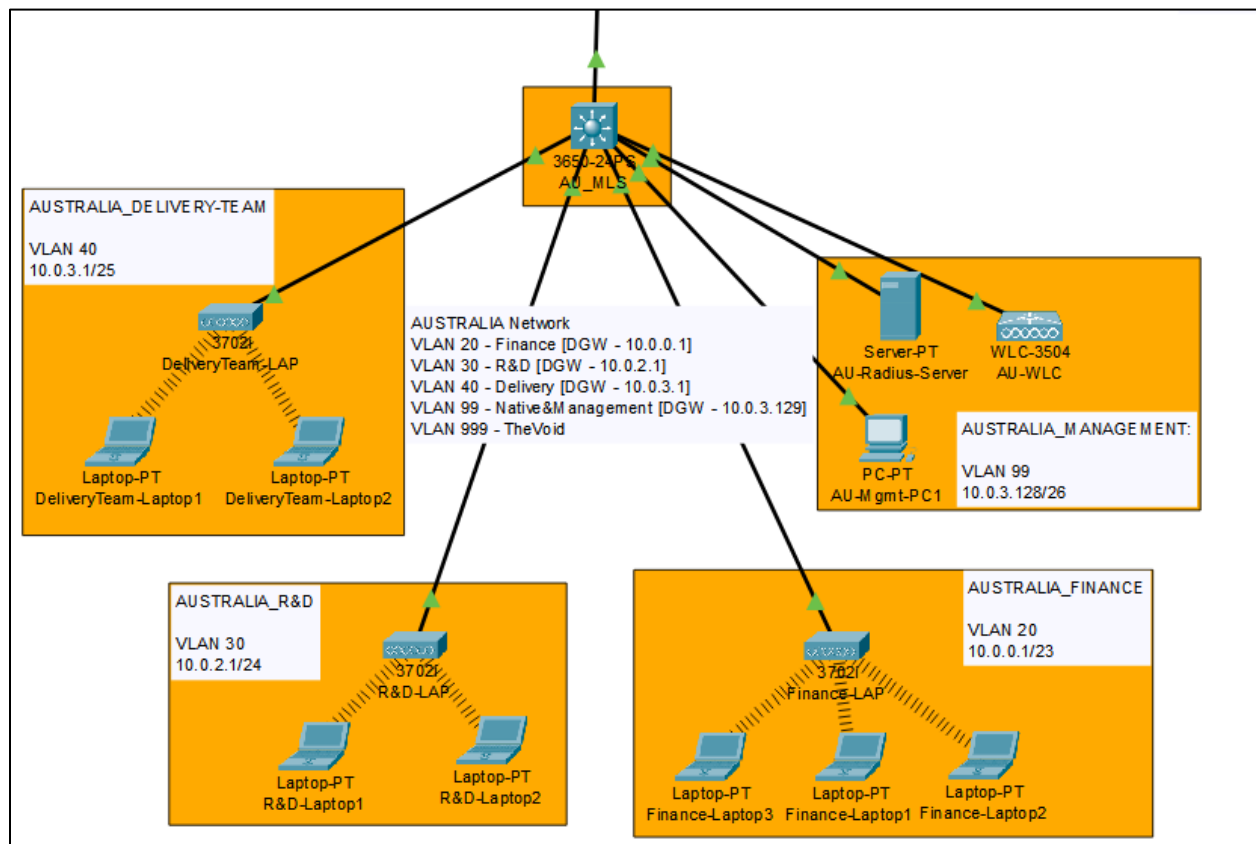


Figure 13 Australia Branch Network Topology

The Australia Branch network architecture is designed around a **Layer 3 Switch (AU_MLS)**, which serves as the core distribution device, providing both high-speed switching and inter-VLAN routing capabilities to efficiently manage network traffic. Unlike traditional wired networks, this branch features a **complete wireless topology** for endpoint connectivity. Access is provided through multiple **Lightweight Access Points (LAPs)** distributed across different departments (Delivery, R&D, and Finance), which are centrally managed by a dedicated **Wireless LAN Controller (WLC)**. To ensure robust security and centralized identity management, the network integrates a **RADIUS AAA Server**, which handles user authentication and authorization before granting access to the wireless network resources.

4.3.2 VLAN Implementation

```

AU_MLS#show vlan brief
VLAN Name                Status    Ports
-----
1    default                active
20   Finance                 active
30   R&D                     active
40   Delivery                active
99   Au-Mgmt&Native          active    Gig1/0/1, Gig1/0/2
999  TheVoid                 active    Gig1/0/3, Gig1/0/4, Gig1/0/5, Gig1/0/6
                                           Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10
                                           Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14
                                           Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18
                                           Gig1/0/19, Gig1/1/1, Gig1/1/2, Gig1/1/3
                                           Gig1/1/4
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
AU_MLS#

```

Figure 14 Australia Branch VLAN Configuration

The Australia branch employs a segmented VLAN strategy to optimize network security and traffic management. The **Management and Native VLAN** functions are consolidated under a single identifier, **VLAN 99** (Au-Mgmt&Native), which isolates administrative traffic and untagged frames from general user data. To further harden the network, a strict port security policy is enforced wherein all unused physical interfaces are administratively shut down and quarantined into a dedicated isolation VLAN, identified as **VLAN 999** or "TheVoid". Consequently, the default **VLAN 1** contains no active ports, ensuring that no traffic traverses the default broadcast domain. Finally, the database lists **VLAN IDs 1002 through 1005**, which are standard legacy VLANs historically reserved for FDDI and Token Ring compatibility.

4.3.3 DHCP Pool Deployment

```

Pool Finance :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 510
Leased addresses : 3
Excluded addresses : 2
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.0.0.1 10.0.0.1 - 10.0.1.254 3 / 2 / 510

Pool R&D :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 2
Excluded addresses : 2
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.0.2.1 10.0.2.1 - 10.0.2.254 2 / 2 / 254

Pool Delivery :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 126
Leased addresses : 2
Excluded addresses : 2
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.0.3.1 10.0.3.1 - 10.0.3.126 2 / 2 / 126

Pool Au-Mgmt&Native :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 62
Leased addresses : 3
Excluded addresses : 2
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.0.3.129 10.0.3.129 - 10.0.3.190 3 / 2 / 62
AU_MLS#

```

Figure 15 Australia Branch IP DHCP Pool Configuration

The Australia Branch network utilizes Variable Length Subnet Masking (VLSM) to efficiently allocate addresses within the 10.0.0.0 block. To automate address assignment, the Multi-Layer Switch functions as the central DHCP server, hosting distinct DHCP pools configured for each department. The Finance Department (VLAN 20) is assigned the largest subnet, **10.0.0.0/23**, utilizing **10.0.0.1** as its default gateway. The R&D Department (VLAN 30) operates on **10.0.2.0/24** with gateway **10.0.2.1**, while the Delivery Team (VLAN 40) uses **10.0.3.0/25** with gateway **10.0.3.1**. Finally, the Management network (VLAN 99) is secured on a smaller **10.0.3.128/26** subnet, utilizing **10.0.3.129** as the gateway.

4.3.4 Interface Trunking and VLAN Tagging

```

AU_MLS#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig1/0/20  on        802.1q         trunking    99
Gig1/0/21  on        802.1q         trunking    99
Gig1/0/22  on        802.1q         trunking    99
Gig1/0/23  on        802.1q         trunking    99

Port      Vlans allowed on trunk
Gig1/0/20 20,30,40,99
Gig1/0/21 20,30,40,99
Gig1/0/22 20,30,40,99
Gig1/0/23 20,30,40,99

Port      Vlans allowed and active in management domain
Gig1/0/20 20,30,40,99
Gig1/0/21 20,30,40,99
Gig1/0/22 20,30,40,99
Gig1/0/23 20,30,40,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig1/0/20 none
Gig1/0/21 none
Gig1/0/22 none
Gig1/0/23 none

AU_MLS#

```

Figure 16 Australia Branch Interface Trunk Configuration

The uplink connectivity for the Australia branch relies on ports **GigabitEthernet1/0/20 through 1/0/23**, which are all configured in **trunking mode** to facilitate the transport of traffic across multiple VLANs. These interfaces utilize the industry-standard **802.1q** encapsulation protocol to tag frames for proper VLAN identification. To align with security best practices, the **Native VLAN** on all trunk ports is set to **VLAN 99**, ensuring untagged traffic is isolated from user data. Furthermore, strict VLAN pruning is enforced; only active service VLANs—specifically **20, 30, 40, and 99**—are allowed to traverse these trunks. Consequently, the default **VLAN 1** and the unused port isolation VLAN **999 ("TheVoid")** are explicitly blocked from the trunk links to minimize the attack surface.

4.3.5 OSPF Dynamic Routing Protocol

```

AU_MLS#show ip protocol

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 5.5.5.5
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.1.255 area 0
    10.0.2.0 0.0.0.255 area 0
    10.0.3.0 0.0.0.127 area 0
    10.0.3.128 0.0.0.63 area 0
    200.100.50.12 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet1/0/1
    GigabitEthernet1/0/2
    GigabitEthernet1/0/20
    GigabitEthernet1/0/21
    GigabitEthernet1/0/22
    GigabitEthernet1/0/23
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:24:46
    2.2.2.2          110          00:02:17
    3.3.3.3          110          00:00:06
    4.4.4.4          110          00:18:55
    5.5.5.5          110          00:18:18
    6.6.6.6          110          00:19:00
    7.7.7.7          110          00:25:47
    8.8.8.8          110          00:25:44
  Distance: (default is 110)

AU_MLS#

```

Figure 17 Australia Branch OSPF Configuration

The Australia Multi-Layer Switch (AU_MLS) utilizes the Open Shortest Path First (OSPF) dynamic routing protocol (**Process ID 10**) to manage network reachability. The device is uniquely identified within the OSPF domain by the **Router ID 5.5.5.5**. It actively advertises four distinct local network subnets, all of which are assigned to the backbone Area 0. To ensure network confidentiality and security, specific interfaces—namely **GigabitEthernet1/0/1 through 1/0/2** and **GigabitEthernet1/0/20 through 1/0/23**—are configured as **passive interfaces**. This configuration prevents the transmission of OSPF routing updates to unauthorized devices or trunk links while still allowing the associated networks to be advertised to the core. Finally, the protocol operates with a default **Administrative Distance of 110**, which determines the trustworthiness of the route source.

4.4 Singapore Branch

4.4.1 Network Topology Overview

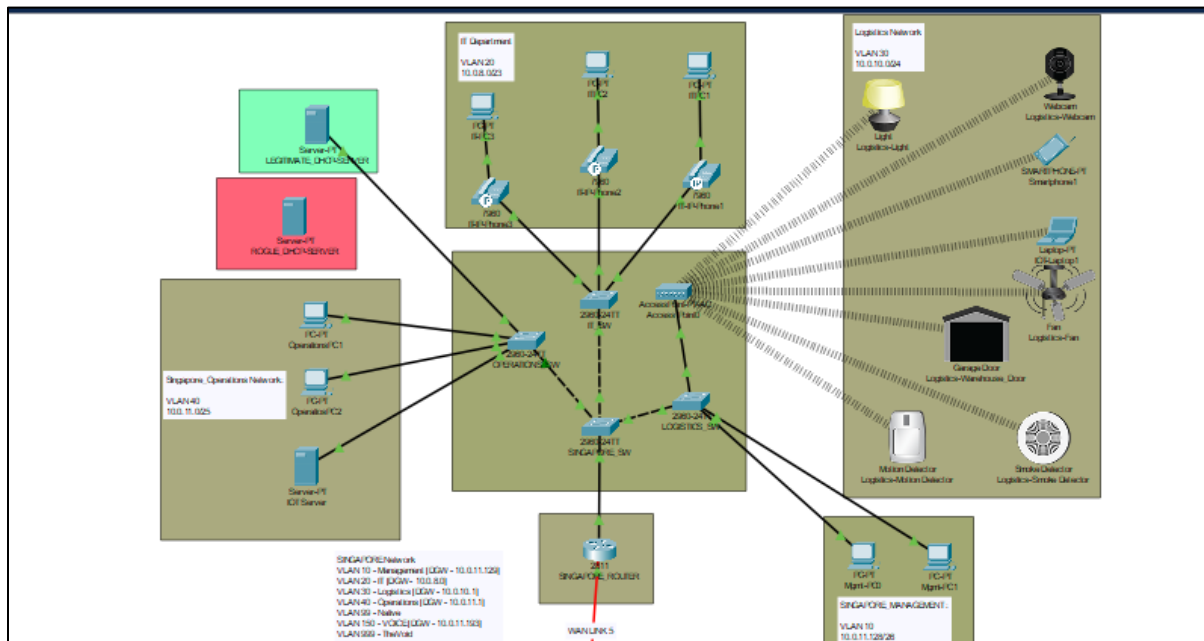


Figure 18 Overview of Singapore Network Topology

The Singapore branch uses a hybrid network design that combines both wired and wireless connectivity to support daily operations. The network is segmented using VLANs to separate IT services, wireless users, and IoT devices, improving efficiency and security. This design allows the Singapore branch to support IT services, wireless operations, and IoT deployment in a controlled and scalable network environment.

Hybrid Wired and Wireless Deployment: The Singapore branch implements partial wireless connectivity using access points to support mobile users and Logistics IoT devices, while critical systems and servers remain on a wired infrastructure. This approach ensures reliable performance for core services while providing flexibility for wireless and IoT operations.

IoT Integration and Network Segmentation: An IoT server is deployed within the Singapore network to centrally manage and monitor IoT devices. Traffic is controlled through VLAN-based segmentation, which enhances security, limits unauthorized access, and allows the network to scale efficiently to support modern enterprise operations.

Voice VLAN Configuration: The Voice VLAN (VLAN 150) is implemented using the telephony-service to support IP telephony within the IT department. Three IP phones are

configured with unique directory numbers: 1010 for IP Phone 1, 2020 for IP Phone 2, and 3030 for IP Phone 3, ensuring proper call routing and isolated voice traffic for improved call quality.

4.4.2 VLAN Implementation

IT_SW:

```
IT_SW#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	Si-Mgmt	active	
20	IT	active	Fa0/1, Fa0/2, Fa0/3
30	Logistics	active	
40	Operations	active	
99	Native	active	
150	VOICE	active	Fa0/1, Fa0/2, Fa0/3
999	TheVoid	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Gig0/1, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
IT_SW#
```

Operations_SW:

```
OPERATIONS_SW#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	Si-Mgmt	active	
20	IT	active	
30	Logistics	active	
40	Operations	active	Fa0/1, Fa0/2, Fa0/3, Gig0/1
99	Native	active	
150	VOICE	active	
999	Thevoid	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
OPERATIONS_SW#
```

Logistics_SW:

```
LOGISTICS_SW#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	Si-Mgmt	active	Fa0/1, Fa0/2
20	IT	active	
30	Logistics	active	Gig0/1
40	Operations	active	
99	Native	active	
150	VOICE	active	
999	Thevoid	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
LOGISTICS_SW#
```

Singapore_SW:

```
SINGAPORE_SW#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	Si-Mgmt	active	
20	IT	active	
30	Logistics	active	
40	Operations	active	
99	Native	active	
150	VOICE	active	
999	TheVoid	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SINGAPORE_SW#
```

Active network traffic across all switches is logically segmented into multiple VLANs to improve traffic organization and enhance security. The design includes VLAN 10 (Management), VLAN 20 (IT), VLAN 30 (Logistics), and VLAN 40 (Operations), which support departmental users and services. A dedicated **Voice VLAN** (VLAN 150) is assigned alongside data VLANs on access ports, ensuring that IP telephony traffic remains isolated from data traffic to maintain voice quality.

For infrastructure and security purposes, VLAN 99 is configured as the native VLAN on trunk links, while VLAN 999 (TheVoid) is used to isolate unused switch ports and prevent unauthorized access. Additionally, VLANs 1002 to 1005 are reserved legacy VLANs defined by Cisco and are not utilized in this network design.

4.4.3 Interface Trunking and VLAN Tagging

IT_SW:

```
IT_SW#Show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on         802.1q         trunking    999

Port      Vlans allowed on trunk
Fa0/24    10,20,30,40,99,150

Port      Vlans allowed and active in management domain
Fa0/24    10,20,30,40,99,150

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    10,20,30,40,99,150

IT_SW#
```

Logistics_SW:

```
LOGISTICS_SW#Show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on         802.1q         trunking    999

Port      Vlans allowed on trunk
Fa0/24    10,20,30,40,99,150

Port      Vlans allowed and active in management domain
Fa0/24    10,20,30,40,99,150

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    10,20,30,40,99,150

LOGISTICS_SW#
```

Operations_SW:

```
OPERATIONS_SW#Show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on         802.1q         trunking    999

Port      Vlans allowed on trunk
Fa0/24    10,20,30,40,99,150

Port      Vlans allowed and active in management domain
Fa0/24    10,20,30,40,99,150

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    10,20,30,40,99,150

OPERATIONS_SW#
```

The show interfaces trunk output verifies that interface Fa0/24 on all three switches (IT_SW, OPERATIONS_SW, and LOGISTICS_SW) is operating correctly as an active 802.1Q trunk link. **VLAN 999** is configured as the **native VLAN**, ensuring that untagged traffic is isolated from user and voice traffic to enhance overall network security.

VLANs 10, 20, 30, 40, 99, and 150 are consistently allowed and active across all trunk links, supporting management, departmental data, and voice traffic between switches. These VLANs are also in the spanning-tree forwarding state and are not pruned, confirming reliable inter-switch connectivity without blocking or misconfiguration.

```

SINGAPORE_SW#Show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    999
Fa0/2     on        802.1q         trunking    999
Fa0/3     on        802.1q         trunking    999
Gig0/1    on        802.1q         trunking    999

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,99,150
Fa0/2     10,20,30,40,99,150
Fa0/3     10,20,30,40,99,150
Gig0/1    10,20,30,40,99,150

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,99,150
Fa0/2     10,20,30,40,99,150
Fa0/3     10,20,30,40,99,150
Gig0/1    10,20,30,40,99,150

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,99,150
Fa0/2     10,20,30,40,99,150
Fa0/3     10,20,30,40,99,150
Gig0/1    10,20,30,40,99,150

SINGAPORE_SW#

```

Figure 19 Singapore Switch Interface Trunk Configuration

The **Singapore switch** functions as the **central connection point** for all departmental VLANs, including **Management, IT, Logistics, Operations, and Voice**. End devices connect to the switch through access ports, where traffic is assigned to the appropriate VLAN to maintain logical separation and secure data flow.

Trunk interfaces on the switch carry VLAN traffic using 802.1Q encapsulation with VLAN 999 as the native VLAN, allowing multiple VLANs to traverse a single link. The **Gig0/1 trunk connects the switch to the Singapore router**, enabling inter-VLAN routing and WAN access so all departments can communicate internally and reach external networks in a controlled and efficient manner.

4.4.4 DHCP Pool Deployment

LEGITIMATE_DHCP-SERVER

Physical Config **SERVICES** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: GigabitEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

Start IP Address: 0.0.0.0

Subnet Mask: 255.255.255.255

Maximum Number of Users: 0

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
VOICE-VLAN150	10.0.11.193	198.81.100.2	10.0.11.203	255.255.255.192	30	10.0.11.193	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	255.255.255.255	0	0.0.0.0	0.0.0.0
SI-MGMT-VLAN10	10.0.11.129	198.81.100.2	10.0.11.139	255.255.255.192	53	0.0.0.0	0.0.0.0
OPERATIONS-VLAN40	10.0.11.1	198.81.100.2	10.0.11.11	255.255.255.128	117	0.0.0.0	0.0.0.0
LOGISTICS-VLAN30	10.0.10.1	198.81.100.2	10.0.10.11	255.255.255.0	245	0.0.0.0	0.0.0.0
IT-VLAN20	10.0.8.1	198.81.100.2	10.0.8.11	255.255.254.0	500	0.0.0.0	0.0.0.0

Figure 20 Singapore DHCP Server Configuration

- ❖ **Management VLAN (VLAN 10):** A dedicated DHCP pool is configured to automatically assign IP addresses to management devices, using the correct default gateway and DNS server to ensure secure and reliable administrative access.
- ❖ **IT VLAN (VLAN 20):** This DHCP pool supports IT department endpoints by providing dynamic IP addressing within the allocated subnet, enabling efficient connectivity for technical operations and system maintenance.
- ❖ **Logistics VLAN (VLAN 30):** A separate DHCP pool is created for Logistics devices, including wired and wireless endpoints, allowing smooth communication while keeping logistics traffic isolated from other departments.
- ❖ **Operations VLAN (VLAN 40):** The Operations VLAN DHCP pool supplies IP addresses to operational users and services, ensuring stable network access while maintaining logical separation from IT and management traffic.

- ❖ **Voice VLAN (VLAN 150):** A dedicated DHCP pool is configured for IP phones, ensuring voice devices receive appropriate IP settings and maintain consistent call quality through isolated voice traffic.

4.4.5 OSPF Dynamic Routing Protocol

```

SINGAPORE_ROUTER#Show IP Protocol

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 6.6.6.6
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.8.0 0.0.1.255 area 0
    10.0.10.0 0.0.0.255 area 0
    10.0.11.0 0.0.0.127 area 0
    10.0.11.128 0.0.0.63 area 0
    10.0.11.192 0.0.0.31 area 0
    200.100.50.16 0.0.0.3 area 0
  Passive Interface(s):
    FastEthernet0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:17:45
    2.2.2.2          110          00:17:47
    3.3.3.3          110          00:17:11
    4.4.4.4          110          00:17:46
    5.5.5.5          110          00:17:12
    6.6.6.6          110          00:17:46
    7.7.7.7          110          00:17:12
    8.8.8.8          110          00:17:11
  Distance: (default is 110)

SINGAPORE_ROUTER#

```

Figure 21 Singapore Router OSPF Configuration

The output confirms that the Singapore router is running OSPF (Open Shortest Path First) **process ID 10**, using the default administrative distance of **110**. The router operates with a single OSPF area (**Area 0**), which serves as the backbone area and ensures efficient and scalable routing across the enterprise network.

Multiple internal networks, including the Management, IT, Logistics, Operations, Voice, and WAN subnets, are advertised into OSPF to enable full inter-VLAN and inter-branch connectivity across the enterprise. The configuration of **FastEthernet0/0 as a passive interface** prevents OSPF routing updates from being sent on end-user facing links, improving security while still allowing the connected network to be advertised.

The routing table displays multiple OSPF neighbors with Router IDs ranging from **1.1.1.1 to 8.8.8.8**, all learned with consistent path costs and recent update timestamps. This confirms stable OSPF adjacencies and reliable route exchange between the Singapore branch and other sites, supporting resilient and dynamic enterprise-wide routing.

4.5 The Internet

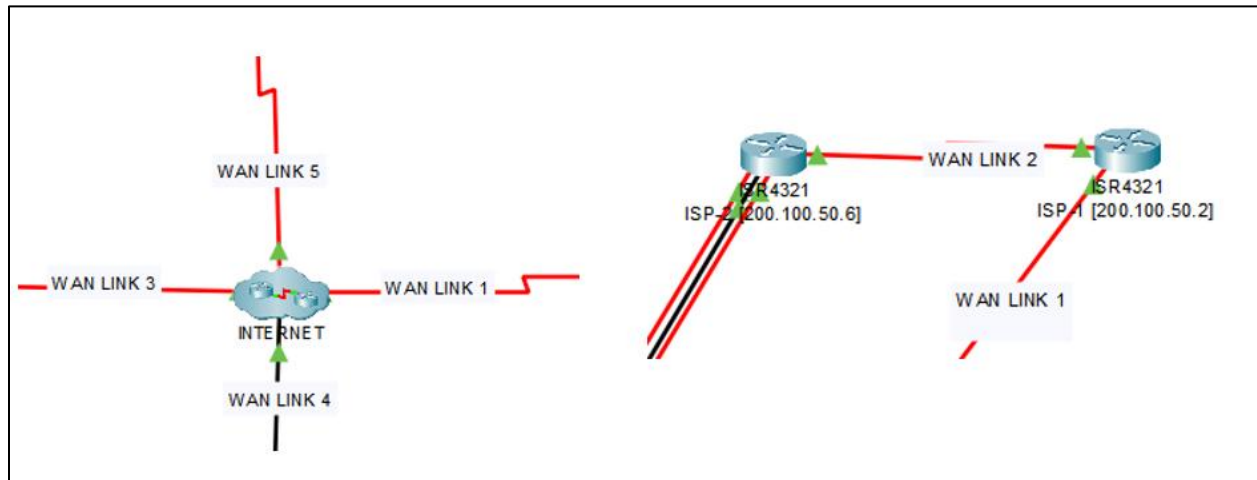


Figure 22 The Internet Cluster formed by ISP Router-1 and ISP Router-2

The network architecture features a robust **Internet Cluster** formed by combining **ISP Router 1** and **ISP Router 2** connected via WAN LINK 2. This cluster serves as the central backbone, interconnecting the four distinct geographical branches: KL-HQ, India, Australia, and Singapore.

ISP Router 1 (KL-HQ Edge): This router is dedicated to servicing the headquarters. It connects directly to the KL-HQ branch via WAN Link 1.

ISP Router 2 (Regional Hub): Functioning as the primary hub for international connections, this router aggregates the WAN links for the remaining branches (India, Australia, and Singapore). These branches connect directly to ISP Router 2, completing the **hub and spoke topology**.

CHAPTER 5: WLAN ARCHITECTURE

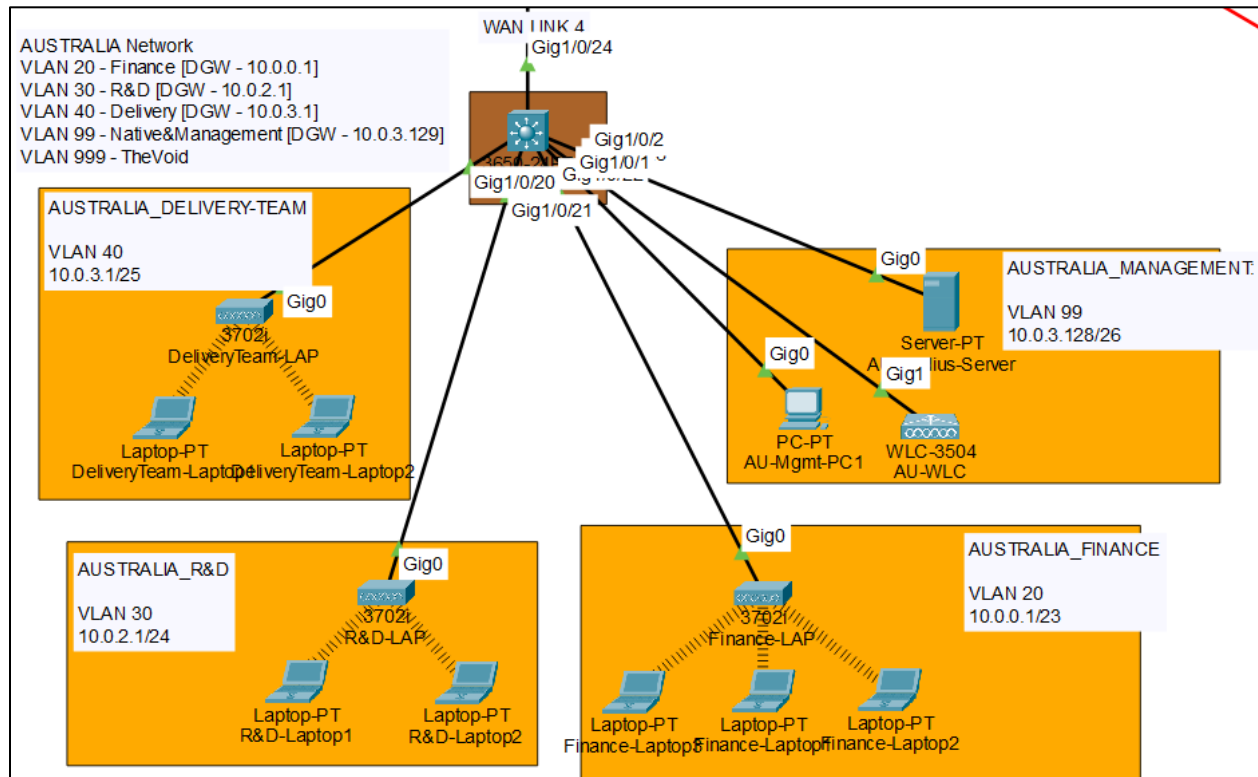


Figure 23 Australia WLAN Architecture

The **Australia branch** uses a **Wireless LAN Controller (WLC)** to centrally manage wireless connectivity. Lightweight Wireless Access Points are deployed to provide wireless access to users across different departments.

Wireless communication is handled through access points controlled by the WLC, ensuring centralized configuration, monitoring, and security enforcement.

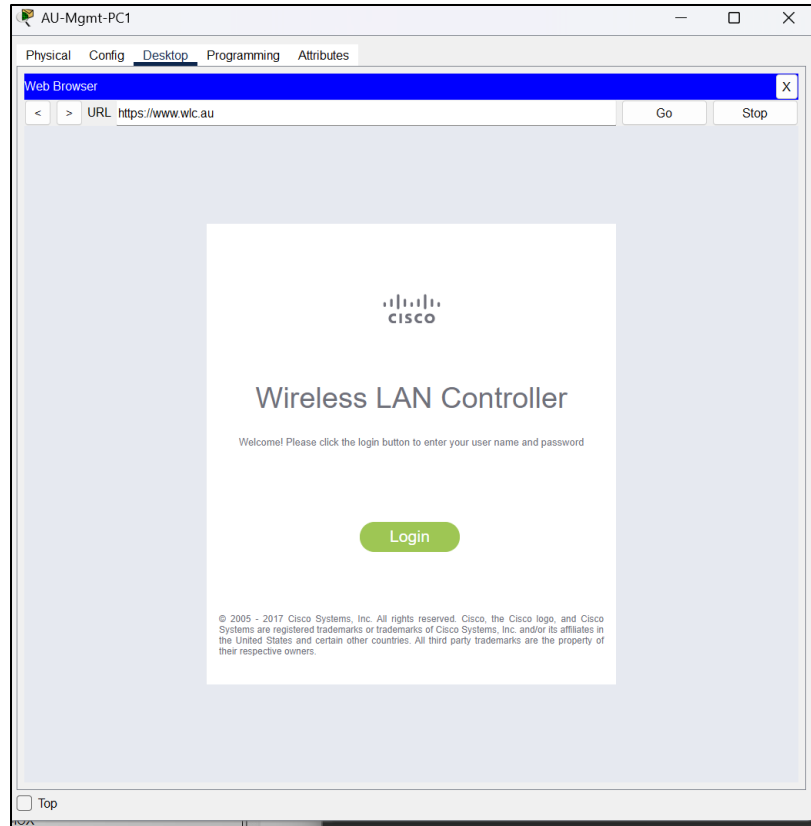


Figure 24 WLC Login Page

The Wireless LAN Controller is **accessed using a web browser** from the management PC (see Figure 24). This interface allows the administrator to monitor and configure wireless settings. The summary page confirms that the controller is active and that access points are successfully connected (see Figure 25).

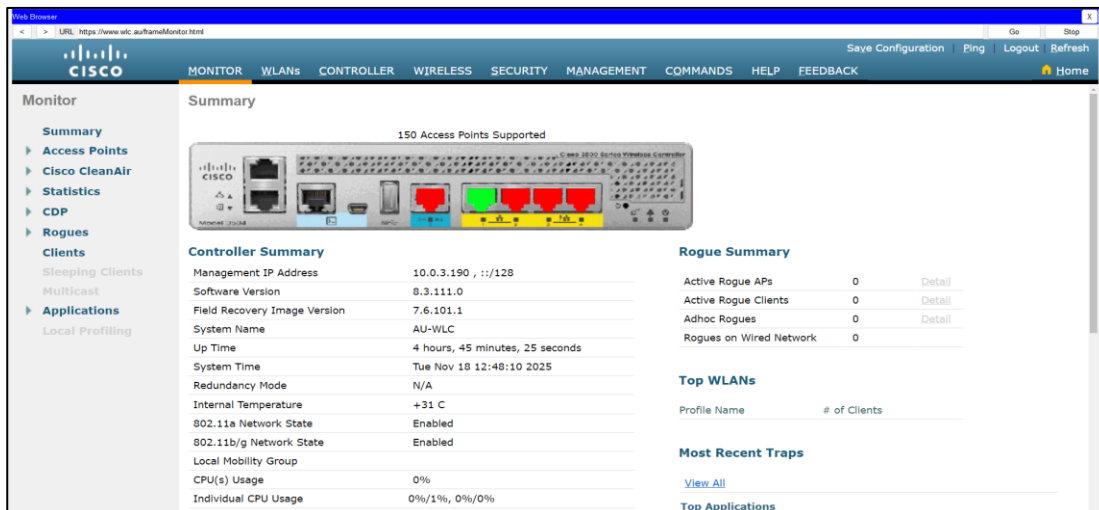


Figure 25 WLC Summary Page

Interface Name	VLAN Identifier	IP Address	Interface Type	Dynamic AP Management	IPv6 Address
Delivery-Wireless	40	10.0.3.5	Dynamic	Disabled	
Finance-Wireless	20	10.0.0.5	Dynamic	Disabled	
R&D-Wireless	30	10.0.2.5	Dynamic	Disabled	
management	1	10.0.3.190	Static	Enabled	::/128
virtual	N/A	192.0.2.1	Static	Not Supported	

Figure 26 WLC Interfaces Overview

The WLC uses different interfaces to separate wireless traffic for each department. Each interface is connected to a different network to keep traffic organized and secure.

WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policies
☐ 1	WLAN	AU_Wireless	AU_Wireless	Enabled	[WPA2][Auth(802.1X)]
☐ 2	WLAN	AU-Finance-Wireless	AU-Finance-Wireless	Enabled	[WPA2][Auth(802.1X)]
☐ 3	WLAN	AU-R&D-Wireless	AU-R&D-Wireless	Enabled	[WPA2][Auth(802.1X)]
☐ 4	WLAN	AU-Delivery-Wireless	AU-Delivery-Wireless	Enabled	[WPA2][Auth(802.1X)]

Figure 27 WLC WLANs Profiles Created

This page shows all wireless networks configured on the controller. Each wireless network is created for a specific department and is managed centrally by the WLC.

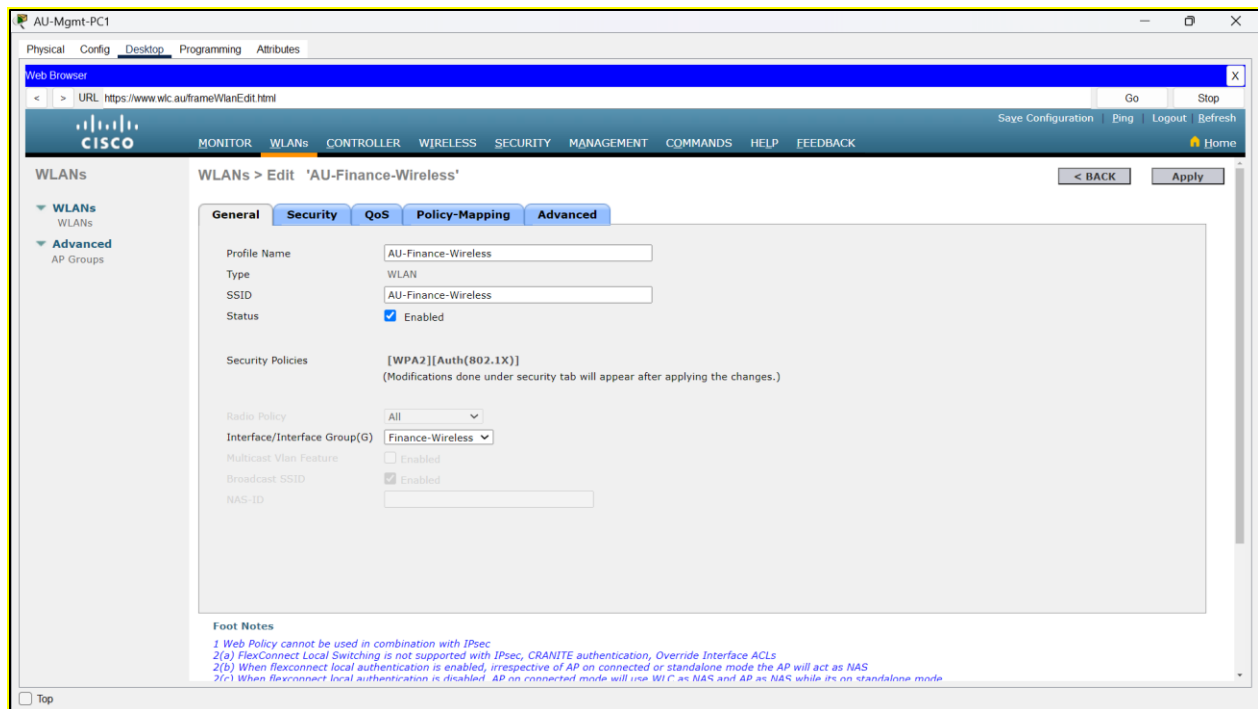


Figure 28 AU-Finance-Wireless WLAN Profile Overview

The selected wireless network is enabled and ready for users to connect. It is linked to the correct network, so users receive proper access when they connect. Each WLAN is mapped to a specific interface to ensure users are placed in the correct departmental network.

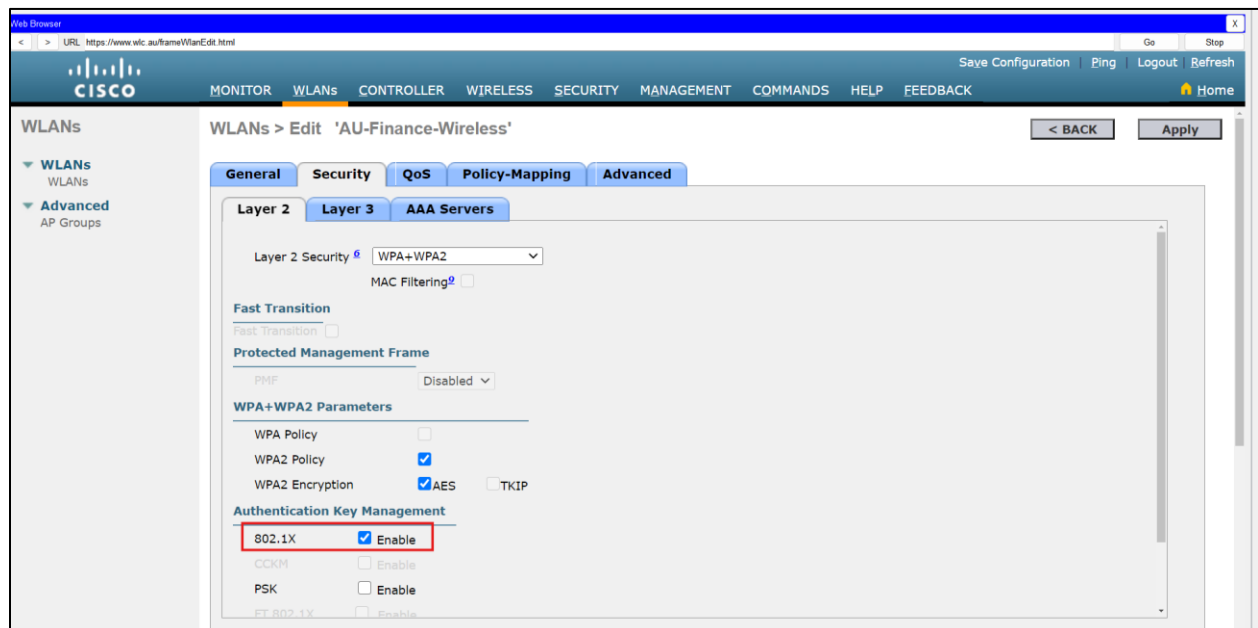


Figure 29 802.1X Protocol Enabled

Wireless security is enabled using WPA2 encryption (see Figure 29). This protects wireless data and prevents unauthorized access to the network.

IEEE 802.1X Authentication

The wireless network uses the IEEE 802.1X authentication protocol. This protocol provides port-based network access control, ensuring that only authenticated users are allowed to connect to the wireless network. When a user attempts to connect, their credentials are verified before network access is granted. This improves security by preventing unauthorized devices from accessing the wireless infrastructure.

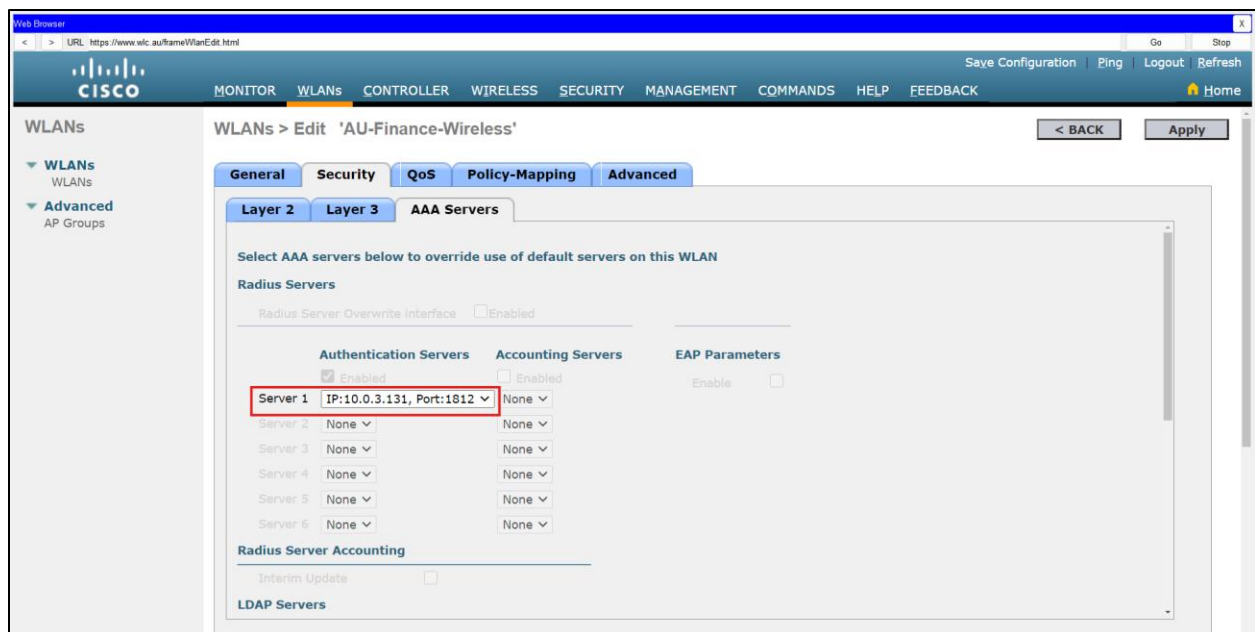


Figure 30 IP Address of RADIUS Server selected along with the Port Number

RADIUS Server Authentication

The Wireless LAN Controller is integrated with a RADIUS server for centralized user authentication. The RADIUS server validates user credentials such as usernames and passwords before granting access to the wireless network. This setup enhances security by ensuring that only authorized users can connect and allows authentication to be managed from a single server.

Wireless Client Connectivity Verification



Figure 31 Wireless Connectivity Verification with WPA2-Enterprise Security

To verify the wireless configuration, a client device was connected using the **PC Wireless** utility. The wireless network status shows a successful connection to the AU-Finance-Wireless SSID with **WPA2-Enterprise security enabled**. This confirms that the Wireless LAN Controller, access points, VLAN configuration, and authentication mechanisms are functioning correctly.

To conclude, the Wireless LAN Controller effectively manages wireless access points, enforces VLAN-based network segmentation, and applies secure authentication using WPA2 and RADIUS. This configuration ensures secure, reliable, and organized wireless connectivity across the Australia branch network.

CHAPTER 6: ETHERCHANNEL

EtherChannel is a **port-link aggregation technology**, combining multiple physical Ethernet links into a single logical one. For example, it could be compared to a big four-lane highway which has no separate one-lane roads that cars are not allowed to move between. Through having, more traffic is able to flow, and even in case one lane gets blocked, the other lanes continue to move.

It raises the total bandwidth of the connection between switches by utilizing all the individual physical ports in the group. If there is a failure of one physical link in the bundle, traffic will be automatically redirected to the other links without network interruption or loop formation. Rather than **Spanning Tree Protocol (STP)** putting redundant links under block to avoid loops, it perceives the whole EtherChannel as a single link, thus making all physical paths active and available.

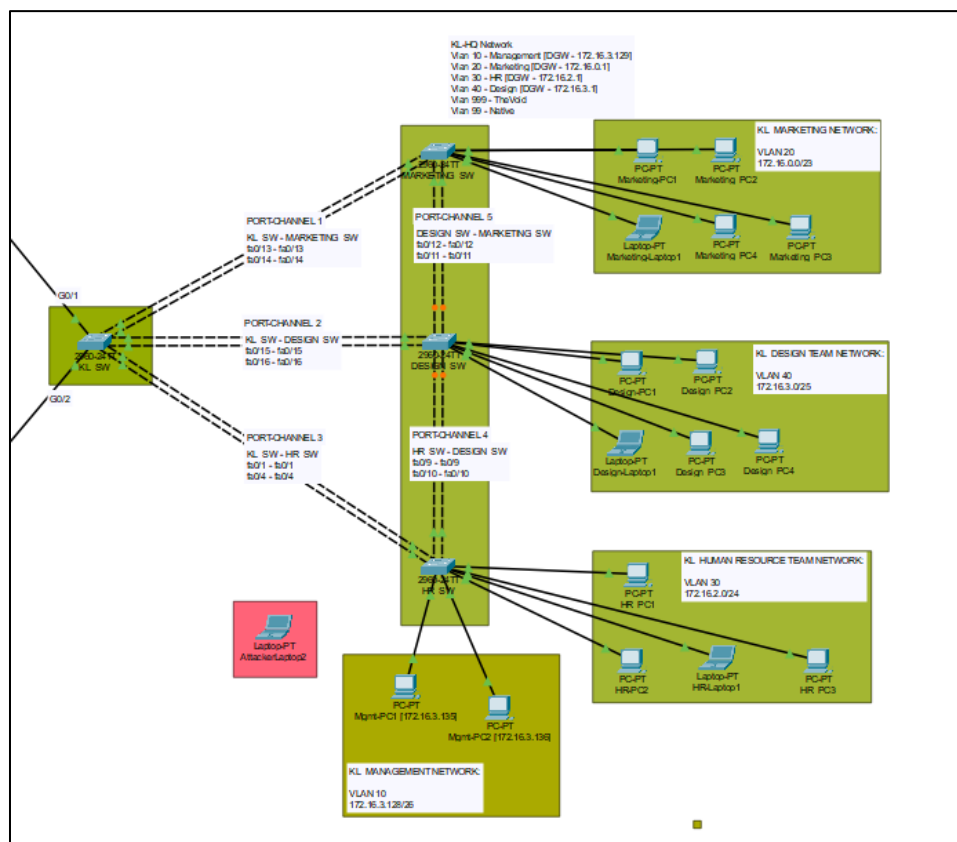


Figure 32 KL EtherChannel Topology

EtherChannel Design

The following EtherChannel links provides link redundancy and bandwidth aggregation between the core and departmental switches:

- PortChannel 1(PO1) : link MARKTING_SW to KL_SE
- PortChannel 2(PO2) : link DESIGN_SW to KL_SW
- PortChannel 3(PO3) : link HR_SW to KL_SW
- PortChannel 4(PO4) : link HR_SW to DESIGN_SW
- PortChannel 5(PO5) : link DESIGN_SW to MARKTING_SW

Link Aggregation Negotiation (LACP)

In the effort to allow the speedy and flexible establishment of EtherChannel's in the KL Headquarters, the implementation of the **Link Aggregation Control Protocol (LACP)**, which is grounded on the **IEEE 802.3ad** standard, was done. LACP is a negotiation protocol that operates automatically, allowing the switches to exchange frames and thus automatically forming a logical Port-Channel. When the choice was made for LACP over static configuration or proprietary protocols, the network was able to guarantee that a bundle would not be formed unless both ends of the link had the same configuration in terms of speed, duplex, and VLAN assignments which in turn eliminates the possibility of network loops or traffic black-holes occurring.

Designated Port (DP)

This is a port in the **Forwarding** state that is responsible for sending traffic down into a network segment. All active Port-Channels on **KL_SW (Core)** are Designated Ports. This ensures the core can always "talk" to the departments.

Root Port (RP)

This is the best path from an Access switch (like Marketing or HR) back to the Core Switch. The primary Port-Channels (Po1, Po2, and Po3) leading to the KL_SW are the **Root Ports** for their respective switches.

Alternate/Blocking Port (Altn/BLK)

A port that is physically connected but "shut off" logically to prevent a bridge loop. Because there are "cross-links" between departments (like **Po5** between Marketing and Design), STP will automatically place one end of that link into a **Blocking** state.

This is good, because If the main cable (Root Port) fails, this "Alternate" port will immediately turn on to keep the department connected.

Verification of EtherChannel Configuration

MARKETING SW: This switch handles the data-intensive tasks of the Marketing department.

```
MARKETING_SW#sh eth sum
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 2
Number of aggregators:          2

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)       LACP       Fa0/13 (P) Fa0/14 (P)
5      Po5 (SU)       LACP       Fa0/11 (P) Fa0/12 (P)
MARKETING_SW#
```

Figure 33 EtherChannel Summary of Marketing Switch

Key Terms

S : Confirms this is a "Switching" bundle (Data Link Layer). It proves its using for VLANs and not as a routed port.

U : This is the most important flag. It means the Port-Channel is **active and forwarding traffic**. If you see "**D**" instead, the **link is down**.

SU : Confirms the entire bundle is healthy and operating at Layer 2(e.g., Po1 (SU)).

P : If this next to physical ports (like Fa0/1(P)), it means that specific wire is successfully "glued" into the bundle.

Port-Channel 1 (Po1) combines **fa0/13 to fa0/13 and fa0/14 to fa0/14**

Port-Channel 5 (Po5) combines **fa0/11 to fa0/11 and fa0/12 to fa0/12**

Po1: This is the main uplink to the **KL_SW** core.

Po5: This is a redundant "cross-link" (likely to the Design switch).

Verification: Both bundles are (SU) and utilize **LACP**. The physical ports Fa0/11 through Fa0/14 are all properly bundled (P), ensuring high availability for marketing media transfers.

KL_SW : This output shows the central "hub" of your EtherChannel configuration. It manages the primary uplinks to three different departmental switches.

```
KL_SW#sh eth sum
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP       Fa0/13(P) Fa0/14(P)
2      Po2(SU)        LACP       Fa0/15(P) Fa0/16(P)
3      Po3(SU)        LACP       Fa0/1(P)  Fa0/4(P)
KL_SW#
```

Figure 34 KL Central Switch EtherChannel Summary

Port-Channel 2 (Po2) combines **fa0/15 to fa0/15 and fa0/16 to fa0/16**

Port-Channel 3 (Po3) combines **fa0/1 to fa0/1 and fa0/4 to fa0/4**

Po1, Po2, and Po3 all show the (SU) flag, confirming they are operational Layer 2 bundles currently in use. They use the **LACP** protocol for dynamic negotiation.

Each channel has two ports marked with the **(P)** flag, proving they are successfully bundled together to increase bandwidth to the departments.

DESIGN_SW: This output is unique because it shows both active and "down" channels, which is great for explaining network logic.

```
DESIGN_SW#sh eth sum
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 5
Number of aggregators:          5

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SD)          -           -
2      Po2 (SU)          LACP        Fa0/15 (P) Fa0/16 (P)
3      Po3 (SD)          -           -
4      Po4 (SU)          LACP        Fa0/9 (P)  Fa0/10 (P)
5      Po5 (SU)          LACP        Fa0/11 (P) Fa0/12 (P)
DESIGN_SW#
```

Figure 35 KL Design Switch EtherChannel Summary

Port-Channel 4 (Po4) combines **fa0/9 to fa0/9 and fa0/10 to fa0/10**

Po2, Po4, and Po5 are all (SU) using **LACP**. These connect the Design department to KL_SW and other departments like Marketing (via Po5).

Po1 and Po3 show the (SD) flag, meaning they are Layer 2 but currently Down.

HR_SW : This switch manages the sensitive data for the Human Resources department.

```

HR_SW#sh eth sum
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 2
Number of aggregators:          2

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
3      Po3 (SU)      LACP       Fa0/1 (P) Fa0/4 (P)
4      Po4 (SU)      LACP       Fa0/9 (P) Fa0/10 (P)
HR_SW#

```

Figure 36 HR Switch EtherChannel Summary

(Po3): The main uplink to the core switch.

(Po4): A secondary link for redundancy

Verification: Both are (SU) with ports **Fa0/1, Fa0/4, Fa0/9, and Fa0/10** successfully bundled (P) using LACP.

CHAPTER 7: HSRP

HSRP (Hot Standby Router Protocol) is a Cisco proprietary redundancy protocol used to provide high network availability. It ensures that if your main router (default gateway) fails, a backup router immediately takes over without breaking network connectivity for users.

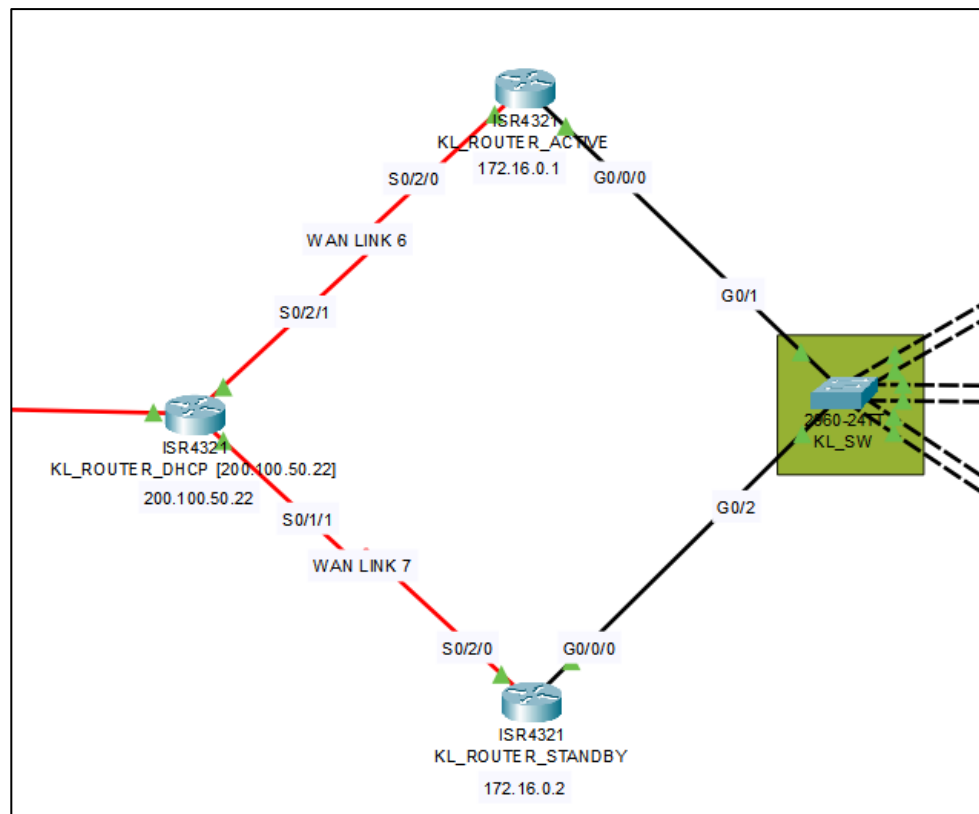


Figure 37 HSRP Implemented in KL-HQ

Figure 31 demonstrates a fault-tolerant network design implemented at the **KL-HQ** site using the **Hot Standby Router Protocol (HSRP)**. The primary objective of this configuration is to ensure high availability and network redundancy; if the primary router fails, the backup router will immediately take over traffic forwarding duties without disrupting connectivity for the LAN users.

Virtual IP Address: Instead of directing end-user devices, such as PCs and phones, to a specific physical router, the network is configured to utilize a **Virtual IP address**. Two or more physical routers collaborate to establish this single "Virtual Router." Users interact exclusively with this Virtual IP, remaining unaware of which specific physical router is actively performing the work.

The HSRP network consists of **three routers** serving distinct roles:

1. KL_ROUTER_ACTIVE (Active Gateway):

- **IP Address:** 172.16.0.1
- **Role:** This is the **primary router** currently handling all traffic from the local network. In normal operations, all data leaving the KL_SW switch passes through this device via the Gigabit Ethernet 0/0/0 interface.

2. KL_ROUTER_STANDBY (Backup Gateway):

- **IP Address:** 172.16.0.2
- **Role:** This router stays in "**Standby**" mode. It monitors the status of the Active router via HSRP hello packets. If the Active router goes down, this device will seamlessly promote itself to active status to maintain network uptime.

3. KL_ROUTER_DHCP (Service Provider):

- **IP Address:** 200.100.50.22
- **Role:** This router acts as the upstream **DHCP server** and WAN connection point. It connects to both the Active and Standby routers via Serial WAN links (WAN LINK 6 and WAN LINK 7), ensuring that whichever router is active has a valid path to the outside network.

HSRP Standby

KL_ROUTER_ACTIVE:

```
KL_ROUTER-ACTIVE#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Gig         1   150 P Active    local         172.16.3.130   172.16.3.190
Gig         1   150 P Active    local         172.16.0.2     172.16.1.254
Gig         1   150 P Active    local         172.16.2.2     172.16.2.254
Gig         1   150 P Active    local         172.16.3.2     172.16.3.126
KL_ROUTER-ACTIVE#
```

Figure 38 KL Active router standby brief overview

- **Priority Configuration:** The **Priority** is set to **150**, which is higher than the default value (100). This higher value manually forces this router to win the election and become the Active gateway.
- **Preemption:** The column **P** is flagged, indicating that **Preemption** is enabled. This ensures that if this router goes offline and later recovers, it will instantly retake the "Active" role from the backup router, restoring the primary traffic path.
- **Virtual IP:** The Virtual IP addresses listed (e.g., 172.16.1.254, 172.16.2.254, see Figure 38) appear to follow a design convention of using the **last usable IP address** of their respective subnets.

KL_ROUTER_STANDBY:

```
KL_ROUTER-STANDBY#sh standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Gig         1   100 Standby    172.16.3.129   local         172.16.3.190
Gig         1   100 Standby    172.16.0.1     local         172.16.1.254
Gig         1   100 Standby    172.16.2.1     local         172.16.2.254
Gig         1   100 Standby    172.16.3.1     local         172.16.3.126
KL_ROUTER-STANDBY#
```

Figure 39 KL Standby-router Standby Brief Overview

- **Priority Configuration:** The **Priority** is set to the default value of **100**. Since 100 is lower than the Active router's 150, this router correctly concedes the election and remains in the background.
- **State Verification:** The **State** is listed as **Standby**, meaning it is currently idle but monitoring the Active router.

CHAPTER 8: SERVERFARM CONFIGURATION

8.1 An Overview

The Server Farm Department in the INDIA branch is responsible for managing essential network services. It operates under VLAN 40 with an IP range of 198.81.100.2 - 198.81.100.5 and includes 4 main servers:

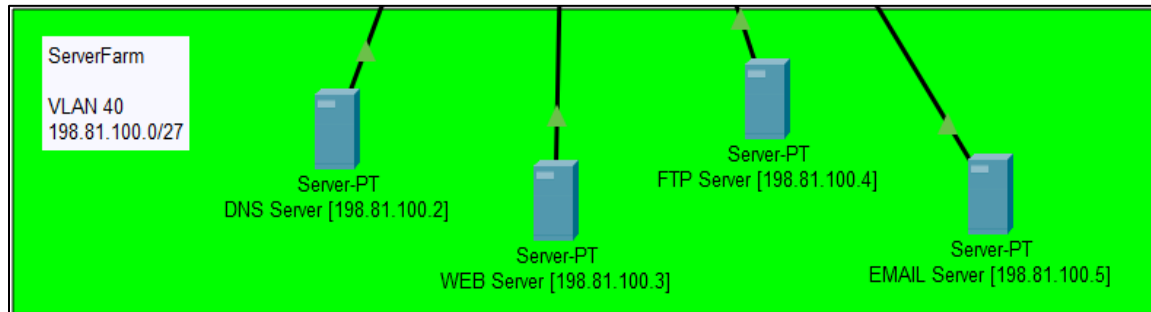


Figure 40 India Server Farm

Key Functions of the India Server Farm

- **DNS Server (198.81.100.2)**
Provides domain name resolution by translating hostnames into IP addresses, allowing users and devices to access internal services efficiently.
- **Web Server (198.81.100.3)**
Hosts internal web pages and web-based applications that can be accessed by users across different VLANs and remote branches.
- **FTP Server (198.81.100.4)**
Enables centralized file storage and file transfer operations, supporting secure upload and download of files within the enterprise network.
- **Email Server (198.81.100.5)**
Facilitates internal email communication by handling the sending and receiving of emails between users within the organization.

8.2 DNS Server

The **DNS server** is installed at India Server Farm where it will offer domain name resolution services to the enterprise network. Its major role is to convert human readable domain names into IP addresses so that a user can access in-house services like **Web, FTP, Email servers** without having to key keyboard the numeric IP addresses. This enhances ease of use, minimizes the configuration mistakes, and centralized network maintenance.

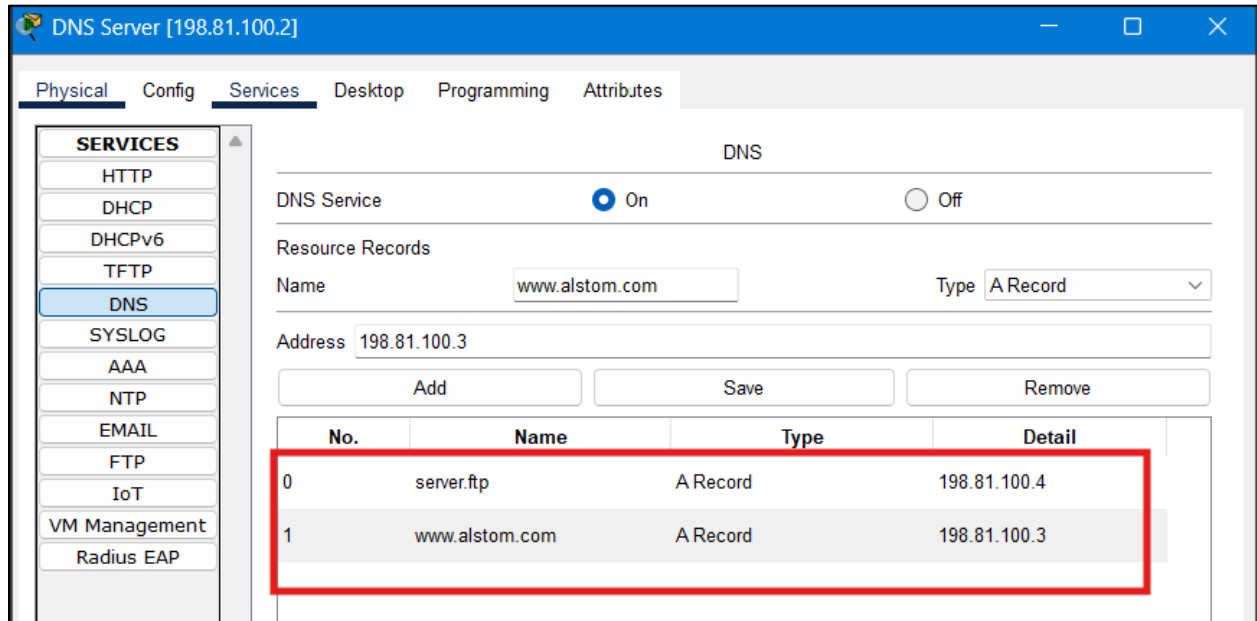


Figure 41 DNS Server Records

The hostname **www.alstom.com** is mapped to the web server IP address **198.81.100.3**, enabling users to access the internal web portal using a domain name instead of an IP address. In addition, the hostname **server.ftp** is mapped to **198.81.100.4**, allowing name-based access to the FTP server.

This configuration confirms that the DNS server is properly set up to support name resolution for multiple internal services, improving usability and ensuring seamless access across different VLANs and remote branches.

8.3 Web Server

The **web server** that was introduced in the **India Server Farm** is to provide web pages that can be accessed by the users across VLANs as well as remote branches hence gives an example of a centralized and web-based information retrieval within the enterprise network.

It can also be checked by accessing it with the help of a web browser by client devices End devices. The fact that the hosted web page is rendered successfully proves that the **HTTP service** works and both inter-VLAN and inter-branch routing are working properly

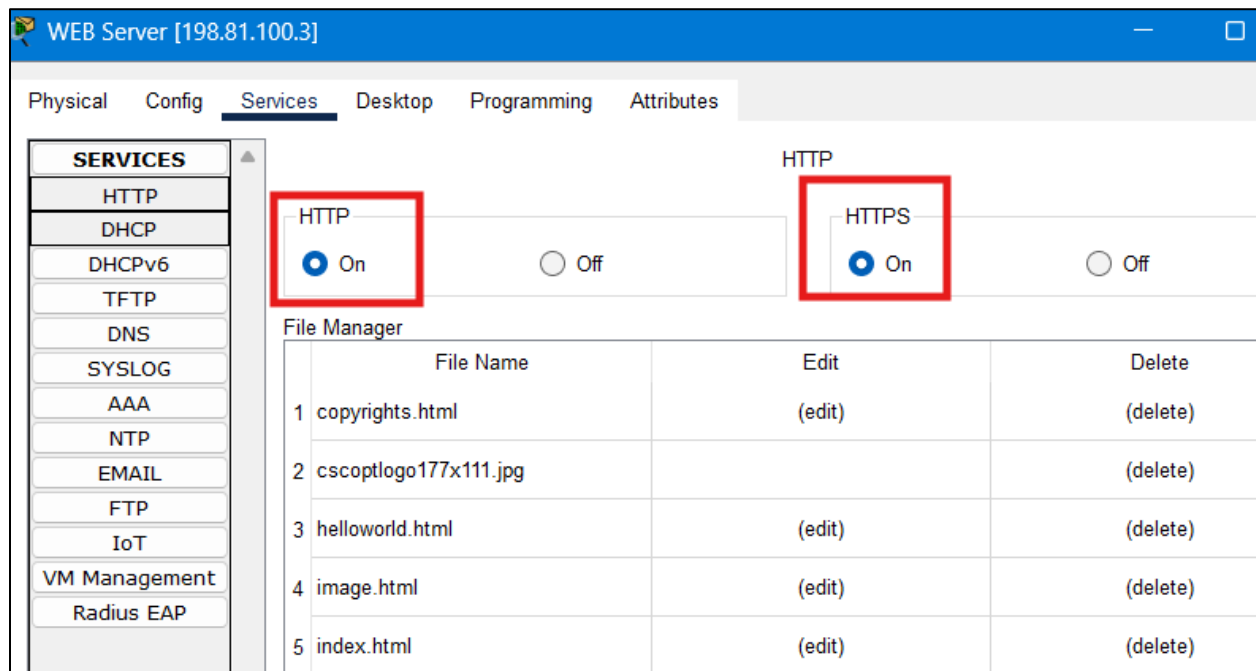


Figure 42 Web server Configuration

The web server is set to run both the **HTTP** and the **HTTPS** services, with the devotes having the ability to use the hosted web content with the standard and safe web protocols.

This is proved by the fact that there are several files of HTML content such as the index.html to show that the server is active to host internal web resources. This architecture shows the effective implementation of the web services in the India Server.

The portrayed web page below affirms that the internal web server is functional, and it is hosting tailor-made content relative to the Alstom India branch.

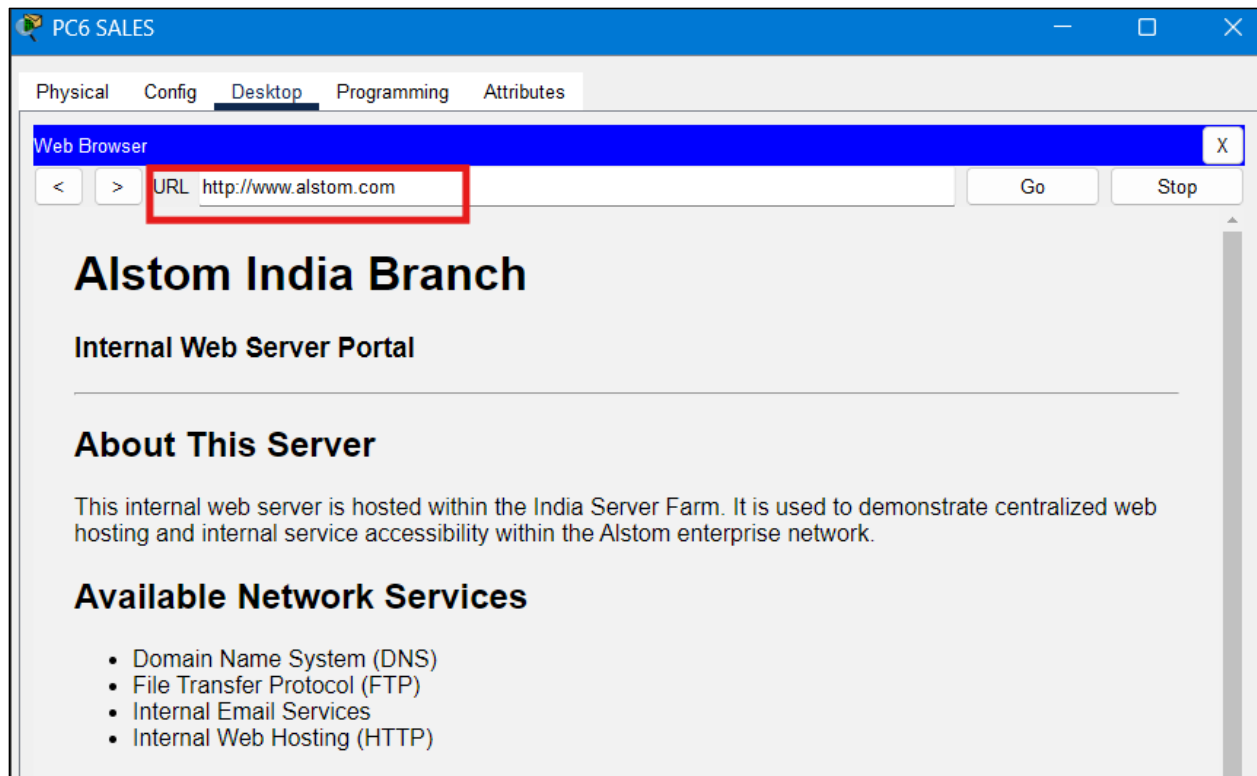


Figure 43 Alstom Company Home Page

When a client device successfully accesses it is also an indication of proper VLAN configuration, inter-VLAN routing and dynamic routing between network segments. This test is an indication that end-to-end connectivity between the user devices and the **Server Farm** is up and running successfully.

8.4 FTP Server

The FTP server is set up in the India Server Farm. This is so that the FTP server can provide a place for people to move files around on the enterprise network. The main job of the FTP server is to let authorized users **upload files download files** and manage files from one place.

Centralizing **file transfer services** within the Server Farm improves network manageability, reduces redundant data storage, and ensures **controlled access to shared resources**. By hosting the FTP server in a dedicated VLAN, server traffic is isolated from user traffic, enhancing both performance and security.

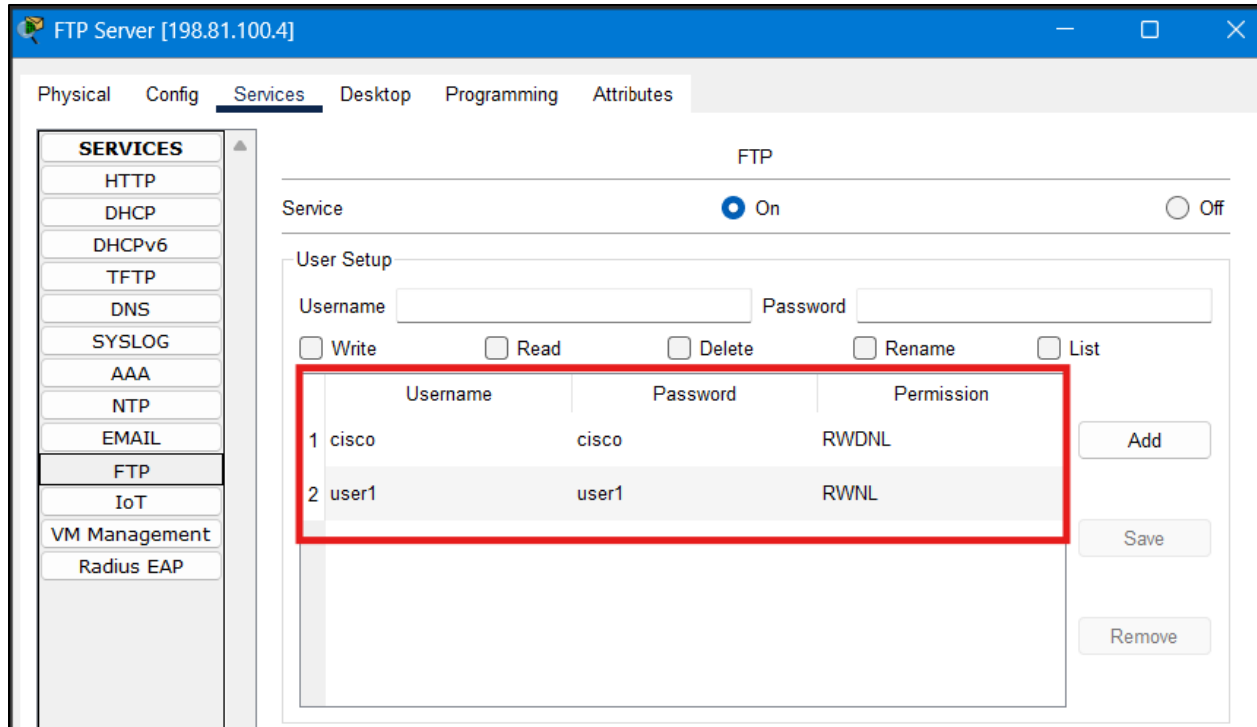


Figure 44 FTP Server Login Credentials

The **FTP server** is set up in the **India Server Farm**. This is being done so that the FTP server can provide a place for file transfer services for the entire enterprise network.

User accounts like **cisco** and **user1** have passwords and special rules about what they can do. They can do things like Read, Write, Delete, Rename and List things. This helps make sure that only the right people can do things with files. It makes everything more secure

The FTP server operates with the IP address **198.81.100.4** and is hosted within the Server Farm VLAN. This placement isolates server traffic from user traffic while allowing controlled access from multiple VLANs and remote branches via inter-VLAN routing and dynamic routing. Successful client login confirms that the FTP service is operational and properly integrated into the network.

A file was created on PC5 Sales and named **ZOZO.txt** (see Figure 45). Then it was uploaded to the FTP server using the FTP protocol. The FTP server and the client device were able to talk to each other. The file was sent successfully.

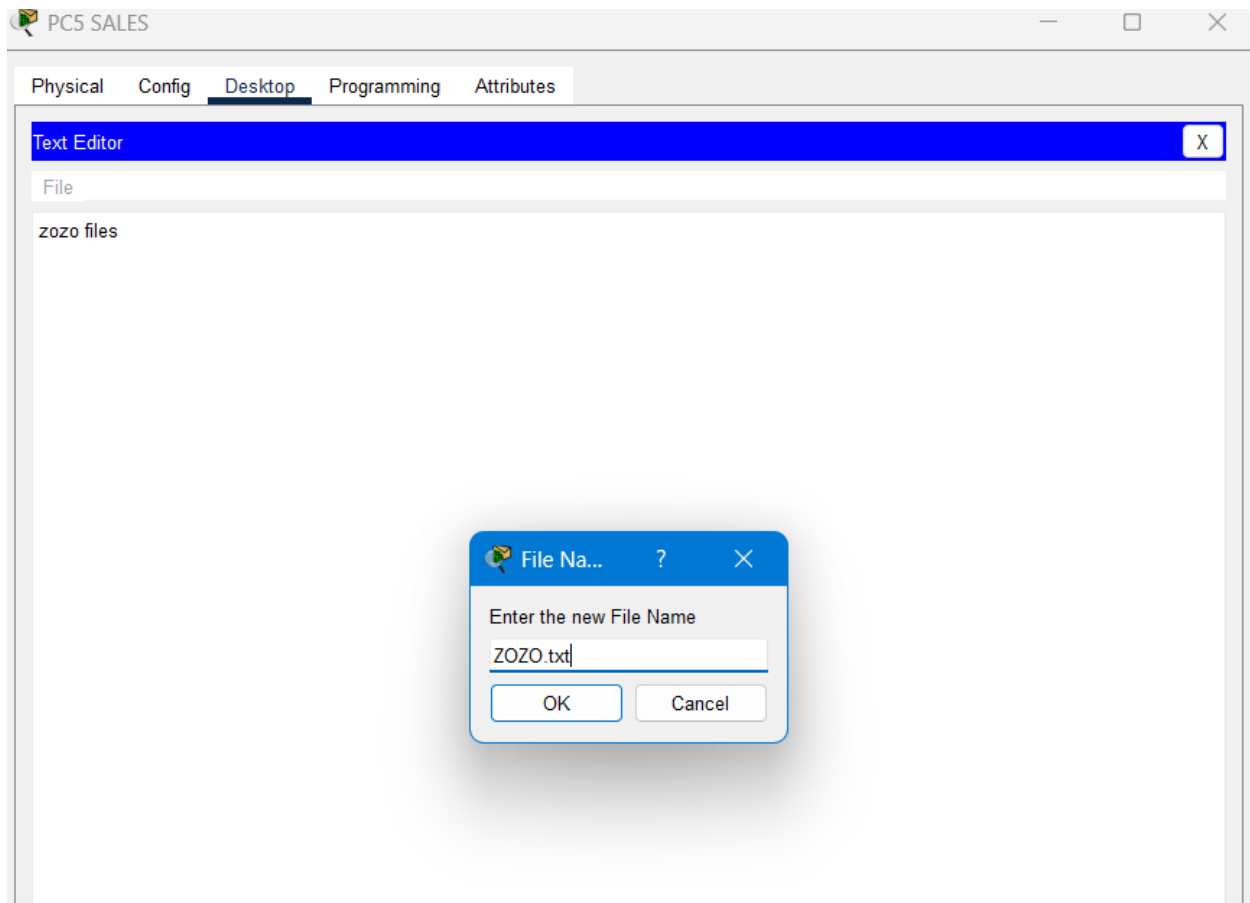


Figure 45 Creating a new text file

The client made a connection to the FTP server on **198.81.100.4**. This connection was successful. The client used user credentials to authenticate. Then the client executed the **put command**. This command was used to upload the file to the **FTP server**. The file was uploaded. This means the FTP server allows writing. It also means the user permissions are set up correctly for the FTP server. The FTP server and the user permissions are working as they should be.

The screenshot shows a Cisco Packet Tracer PC Command Line window for a PC named 'PC5 SALES'. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt is open, showing the following commands and output:

```

Cisco Packet Tracer PC Command Line 1.0
C:\>dir

Volume in drive C has no label.
Volume Serial Number is 5E12-4AF3
Directory of C:\

1/1/1970   8:0 PM           9      ZOZO.txt
1/1/1970   8:0 PM          26     samplefile.txt
               35 bytes          2 File(s)

C:\>ls
Invalid Command.

C:\>ftp 198.81.100.4
Trying to connect...198.81.100.4
Connected to 198.81.100.4
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(/passive mode On)
ftp>put ZOZO.txt

Writing file ZOZO.txt to 198.81.100.4:
File transfer in progress...

[Transfer complete - 9 bytes]

9 bytes copied in 0.166 secs (54 bytes/sec)
ftp>dir

Listing /ftp directory from 198.81.100.4:
0   : ZOZO.txt                               9
1   : asa922-k9.bin                          3371984
2   : asa923-k9.bin                          30468096
3   : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
4   : c1841-ipbase-mz.123-14.T7.bin           13832032
5   : c1841-ipbasek9-mz.124-12.bin            16599160
6   : c1900-universalk9-mz.SPA.155-3.M4a.bin  33591768
7   : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
8   : c2600-i-mz.122-28.bin                   5571584
9   : c2600-ipbasek9-mz.124-8.bin             13169700
10  : c2600-advipservicesk9-mz.124-15.T1.bin 5000000
  
```

Figure 46 Logging into FTP via the Command Prompt

Subsequent execution of the **dir** command lists the uploaded **ZOZO.txt** file within the **FTP server directory**, verifying that the file transfer was completed successfully (see Figure 46). This confirms proper FTP service configuration, user access control, and end-to-end network connectivity between the client VLAN and the Server Farm in India Branch.

8.5 Email Server

The **Email server** is configured within the **India Server Farm** to provide internal email communication services for the enterprise network. As shown in the configuration, both the **SMTP** and **POP3** services are enabled. **SMTP** is responsible for sending outgoing emails, while **POP3** allows users to retrieve incoming emails from the server.

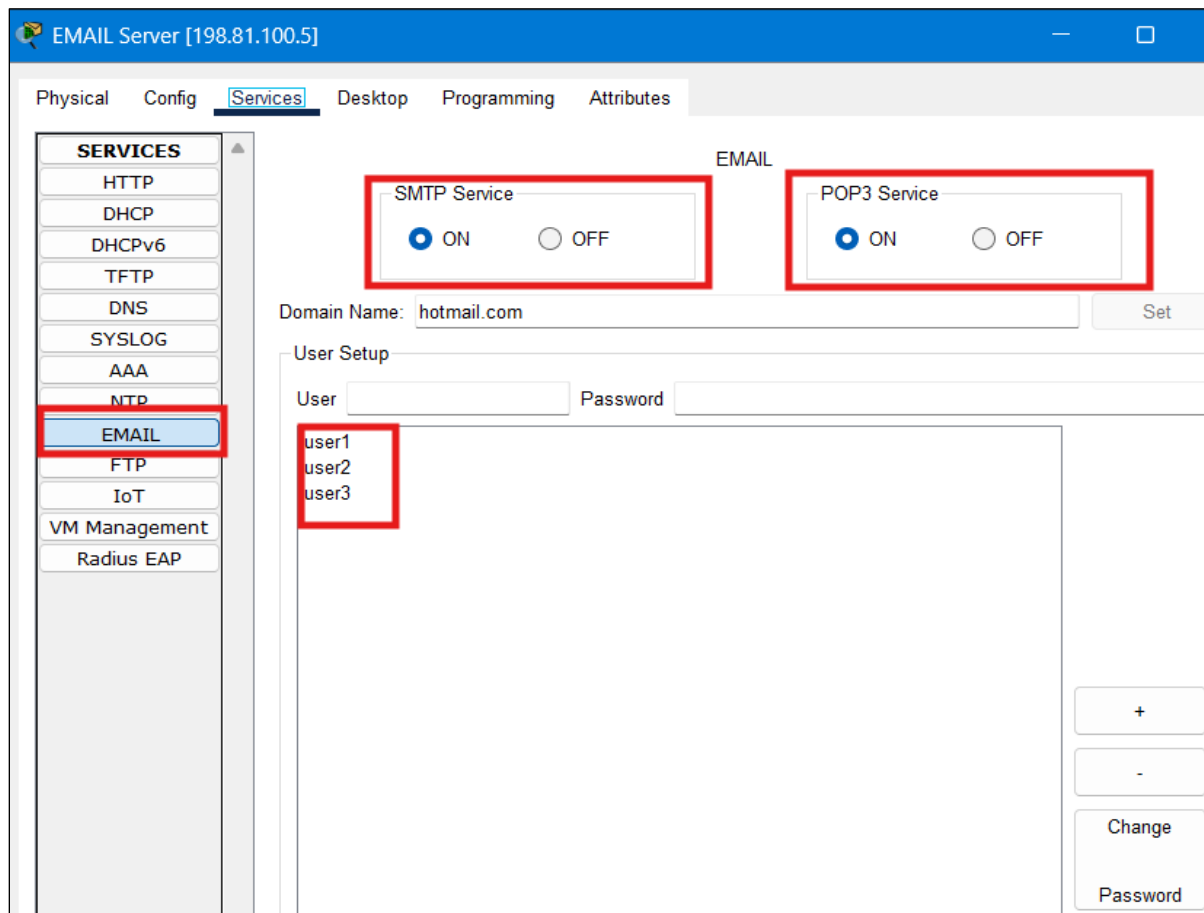


Figure 47 Email Server Credentials and User Setup

The **Email server** works with the internet address **198.81.100.5**. It is connected to the domain name **hotmail.com**. This Email server and the domain name hotmail.com are used to send emails to people inside the network.

The Email server has user accounts (see Figure 47) on it so that people in different parts of the company can send emails to each other. This helps the Email server and the people who use Email to talk to each other.

Enabling both **SMTP** and **POP3** services confirms that the Email server can handle complete email transactions, including message transmission and retrieval. Successful email exchange between configured user accounts verifies that the Email server is operational and properly integrated with VLAN configuration, inter-VLAN routing, and dynamic routing across the network.

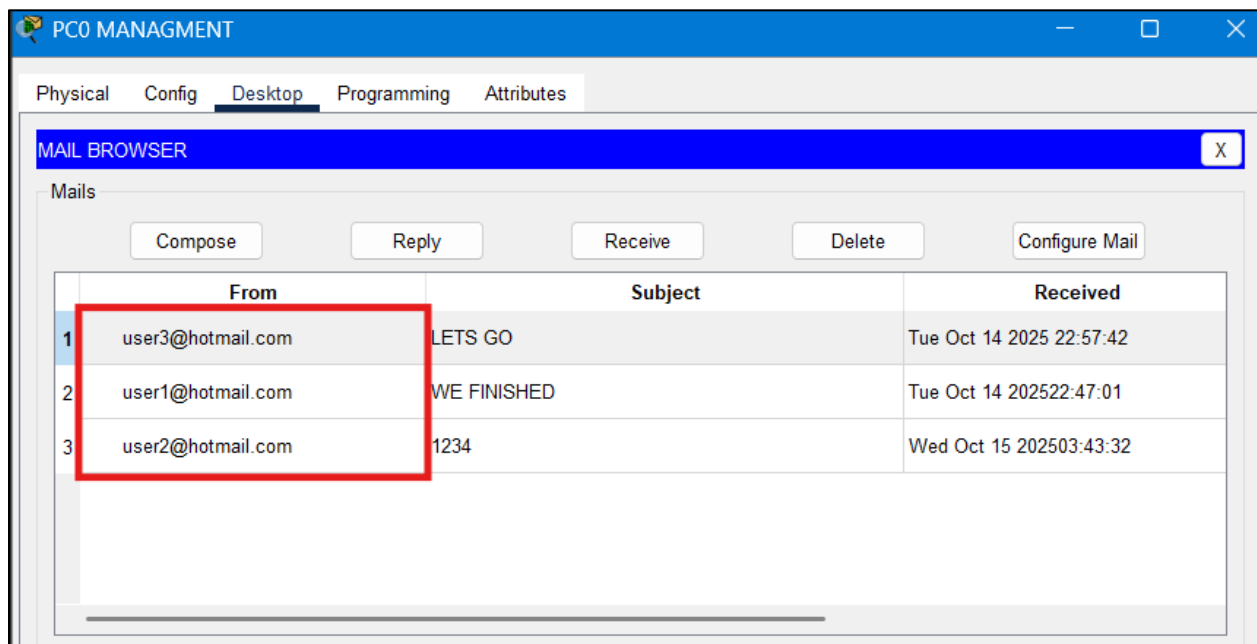


Figure 48 The Recipient Successfully received Emails

The presence of received messages indicates that both **SMTP** and **POP3** services configured on the Email server are operational. **SMTP** is used to send outgoing emails, while **POP3** enables users to retrieve incoming messages from the server. Successful email delivery and retrieval confirm that the Email server is reachable across VLANs and correctly integrated with inter-VLAN routing and dynamic routing.

Overall, this verification proves that the **Email server** hosted in the **India Server Farm** is fully functional and capable of supporting reliable internal communication within the enterprise network.

CHAPTER 9: TRACERT/ PING COMMANDS

This section verifies the successful interconnection of networks within the Alstom infrastructure. The use of **ping** and **tracert** commands is intended to demonstrate reliable communication between different network segments. Analysis of the traceroute paths and ping responses allows network administrators to confirm the effectiveness of the implemented configurations and ensure efficient data transmission across all interconnected network components.

9.1 KL-HQ To Singapore

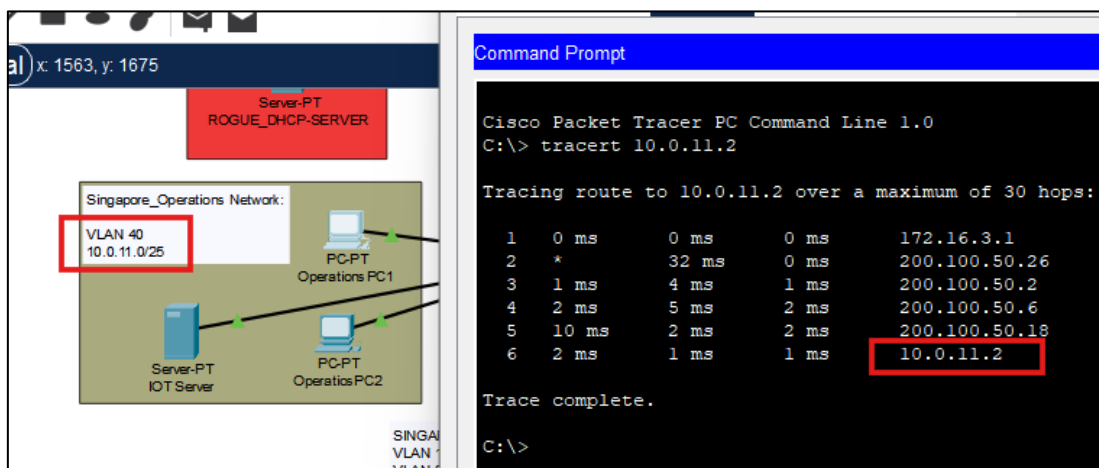


Figure 49 Tracert Successfully completed from KL-HQ to Singapore Branch

9.2 Singapore to KL-HQ

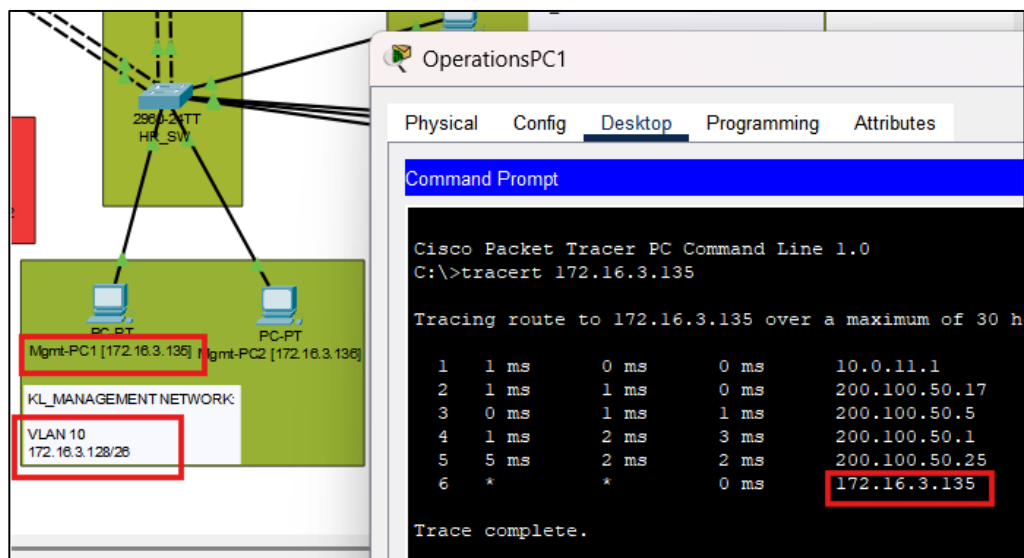


Figure 50 Tracert Successfully Completed from Singapore Branch to KL-HQ

9.3 India to Australia

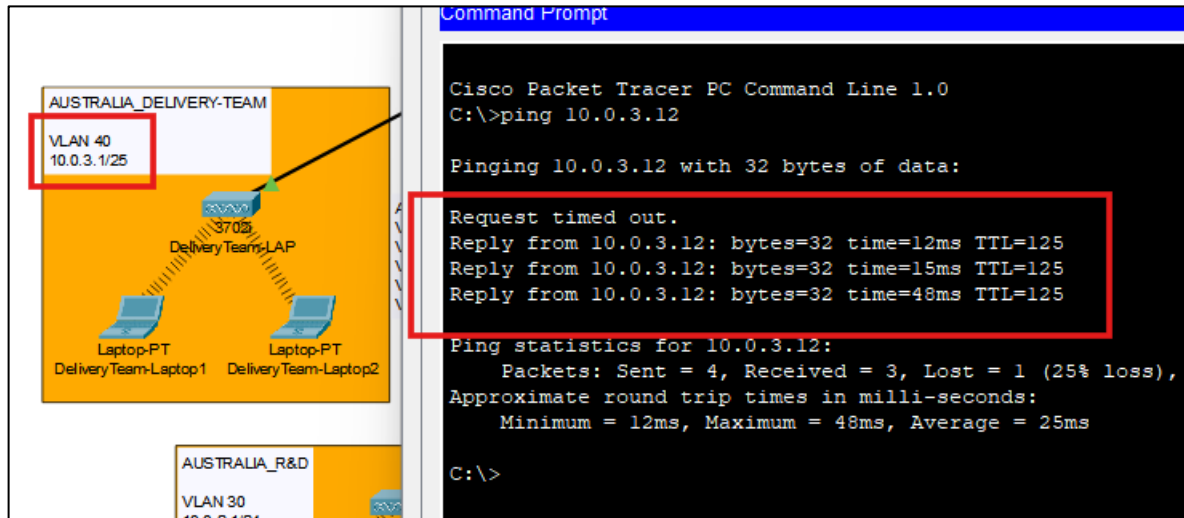


Figure 51 Successful Ping from India Branch to Australia Branch

9.4 Australia to India

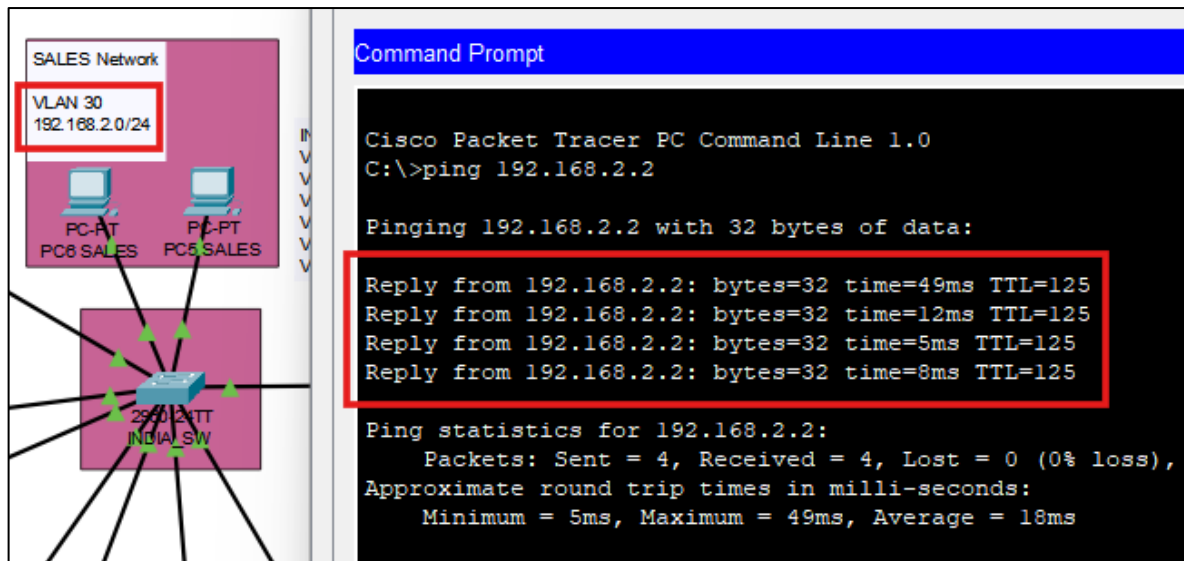


Figure 52 Successful ping from Australia Branch to India Branch

9.5 KL-HQ to India

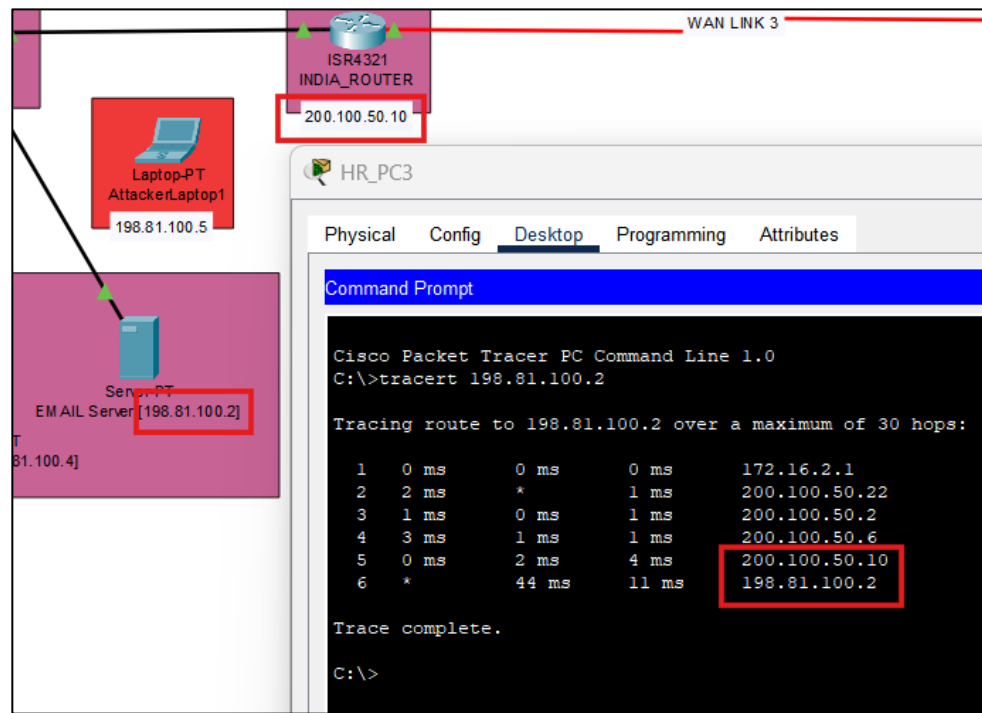


Figure 53 Tracert Successfully completed from KL-HQ to India Branch

9.6 India to KL-HQ

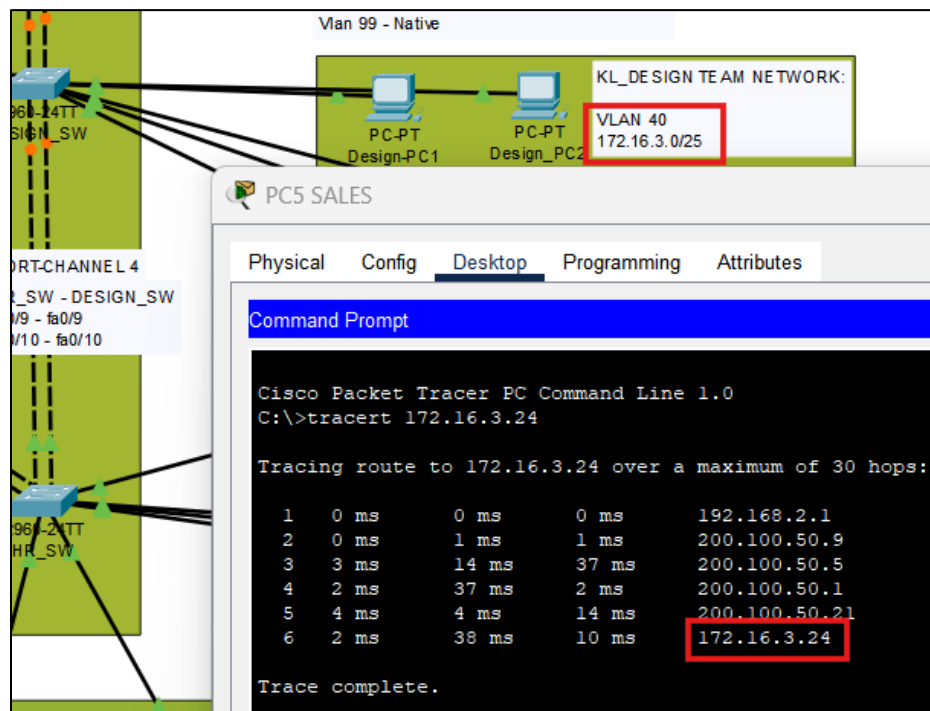


Figure 54 Tracert Successfully completed from India to KL-HQ

9.7 Singapore to Australia

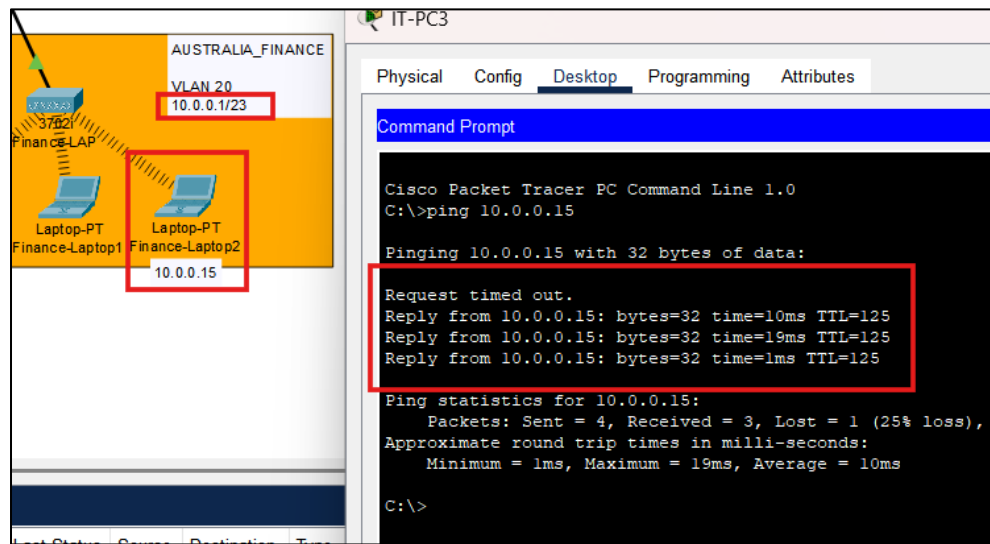


Figure 55 Successful ping from Singapore Branch to Australia Branch

9.8 Australia to Singapore

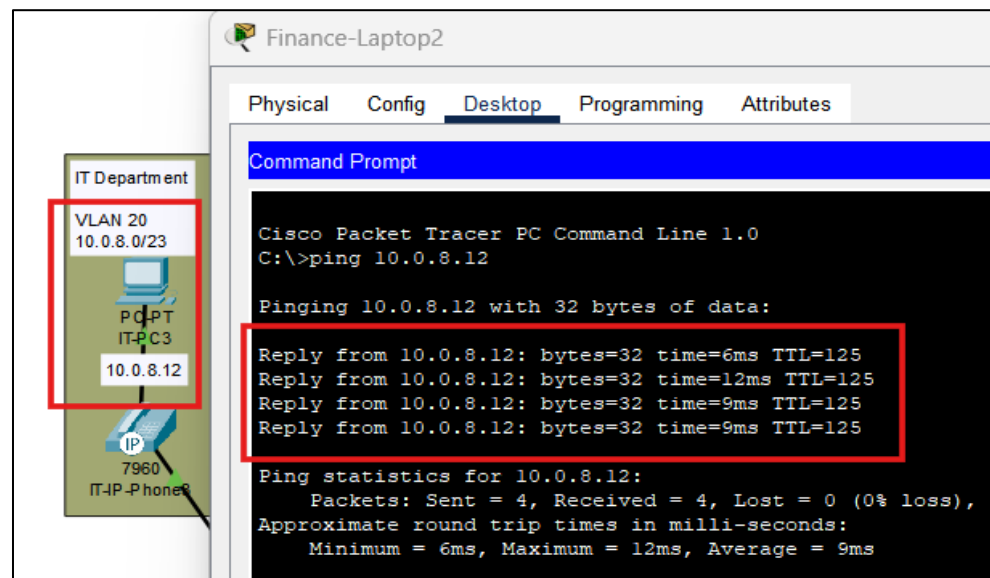


Figure 56 Successful ping from Australia Branch to Singapore Branch

In conclusion, the successful results obtained from the *ping* and *tracert* commands confirm that all network segments within the Alstom infrastructure are properly interconnected. The absence of packet loss and the expected routing paths demonstrate that the routing, switching, and security configurations are functioning as intended. This validation confirms reliable end-to-end connectivity and efficient data transmission across the enterprise network.

CHAPTER 10: SECURITY MECHANISMS

10.1 MAC Address Table Attack

How it works: A switch maintains a "CAM Table" (Content Addressable Memory) that maps MAC addresses to specific physical ports. This table has a limited size. In this attack, a hacker uses a tool to generate hundreds of thousands of fake MAC addresses and floods the switch with them.

Port Security is a Cisco IOS feature that stops this attack by limiting how many MAC addresses are allowed on a single port. If a port is meant for one user, it should only ever see one MAC address.

```
INDIA_SW#show port-security
```

Secure Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
Fa0/5	1	1	0	Shutdown
Fa0/6	1	1	0	Shutdown
Fa0/7	1	1	0	Shutdown
Fa0/8	1	1	0	Shutdown

```
INDIA_SW#
```

Figure 57 Port-Security enabled on Server Farm Ports of India Switch

India ServerFarm Security: Port security is enabled on the **ServerFarm** ports **Fa0/5**, **Fa0/6**, **Fa0/7**, and **Fa0/8** of the India Switch. This confirms these ports are actively monitoring for unauthorized connections.

Strict Access Control: The **MaxSecureAddr** (Maximum Secure Address) is **set to 1**. This means each port will only allow exactly one specific device (MAC address) to communicate. If a user tries to plug in a second device or a hub, it will trigger a violation.

Critical Defense Mode: The **Security Action** is configured as **Shutdown**. Since the ServerFarm is a critical and sensitive location, this is the most secure option. If any violation occurs (like an intruder trying to connect a rogue laptop), the switch will immediately disable the interface to protect the data, requiring an administrator to manually restore it.

10.2 DHCP Spoofing Attack

How it works: The attacker broadcasts thousands of fake DHCP requests with different spoofed MAC addresses. The DHCP server thinks these are real computers and assigns them IP addresses until its entire pool of IP addresses is empty. Real users can no longer get an IP or access the network.

DHCP Snooping acts as a firewall between untrusted hosts and legitimate DHCP servers. It allows the switch to distinguish between ports that are allowed to *send* DHCP offers and those that are not. It classifies every port on the switch into two categories:

1. **Trusted Ports:** These are ports connected to legitimate DHCP Servers or Uplinks (trunks) leading to them. The switch **allows** DHCP Offers to pass through these ports.
2. **Untrusted Ports:** These are ports connected to standard users (PCs, Printers). The switch **blocks** any DHCP Server messages (Offers/Acks) coming from these ports.

```

OPERATIONS_SW#show IP DHCP Snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
10,20,30,40,150
Insertion of option 82 is disabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                Trusted    Rate limit (pps)
-----
FastEthernet0/2          no        15
FastEthernet0/1          no        15
FastEthernet0/24         no        15
GigabitEthernet0/1       yes       unlimited
OPERATIONS_SW#

```

Figure 58 IP DHCP Snooping enabled on Operations Switch of Singapore Branch

IP DHCP Snooping is enabled globally on the **Operations Switch** (OPERATIONS_SW) of the Singapore Branch. This security feature is currently active and monitoring DHCP messages to prevent rogue DHCP servers and untrusted traffic. A **security rate limit 15** has been applied to the untrusted ports to prevent DHCP starvation or flooding attacks. DHCP Snooping is explicitly configured and active on **VLANs 10, 20, 30, 40, and 150**.

10.3 VLAN Hopping Attack

How it works: The attacker configures their computer to pretend it is a switch. It negotiates a "trunk" link with the real switch. The attacker then sends a data frame with **two** 802.1Q VLAN tags attached to it. The first switch strips off the first tag and forwards the packet. The second switch sees the second tag and forwards the packet to the victim's restricted VLAN. This bypasses network segmentation, allowing unauthorized access to sensitive or restricted network segments.

Disable Dynamic Trunking Protocol (DTP) and force the port to be an Access port (for users) or a Trunk port (for real switches). Do not use "Dynamic".

```
AU_MLS#sh interface g1/0/20 switchport
Name: Gig1/0/20
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 99 (Au-Mgmt&Native)
Voice VLAN: none
```

Figure 59 DTP is Disabled on Australia MLS Switch Trunk port Gig1/0/20

To neutralize this threat, a defense-in-depth strategy known as a "**Black Hole**" or "**Parking Lot**" VLAN is implemented. This technique involves creating a dedicated VLAN specifically for unused ports.

```
AU_MLS#sh vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
20	Finance	active	
30	R&D	active	
40	Delivery	active	
99	Au-Mgmt&Native	active	Gig1/0/1, Gig1/0/2
999	TheVoid	active	Gig1/0/3, Gig1/0/4, Gig1/0/5, Gig1/0/6 Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10 Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14 Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18 Gig1/0/19, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
AU_MLS#
```

Figure 60 All the unused ports of AU_MLS are placed in "TheVoid" VLAN

10.4 SSH

To enable SSH, the device must have a unique identity. We define this using a hostname and a domain name. Hostname is AU_MLS

```
AU_MLS#conf t
Enter configuration commands, one per line. End with CNTL/Z.
AU_MLS(config)#username administrator secret cisco
AU_MLS(config)#ip domain-name natacad.pka
AU_MLS(config)#crypto key generate
AU_MLS(config)#crypto key generate rsa

AU_MLS(config)#line vty 0 15
*Mar 1 2:11:34.877: %SSH-5-ENABLED: SSH 1.99 has been enabled
AU_MLS(config-line)#login local
AU_MLS(config-line)#transport input ssh
AU_MLS(config-line)#exit
AU_MLS(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
AU_MLS(config)#
```

Preparation:

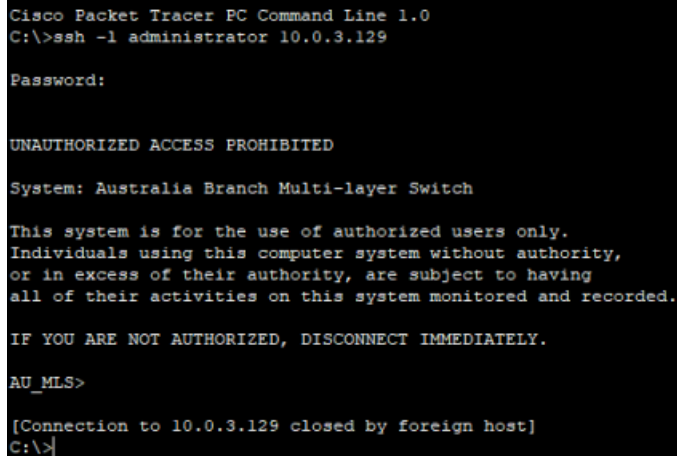
- **username administrator secret cisco:** The local user account "administrator" is created along with the password "cisco". The secret keyword makes sure that the password is only readable in an encrypted form in the configuration file.
- **ip domain-name natacad.pka:** This command assigns a DNS domain name. It is the first step toward encryption key generation; the device uses its hostname and the domain name to create its identity.
- **crypto key generate rsa:** The command issues an ultimatum to produce the corresponding RSA key pair that will be utilized to encipher the SSH connection. The SSH process is completely dependent on these keys to initiate.

Activation & Security:

- **line vty 0 15:** The configuration of the VTY lines for Virtual Teletype access is being done. The numbers 0 15 indicate that up to 16 users can be connected to the system remotely at the same time.
- **login local:** The switch will utilize the local database for authentication whenever a login attempt is made.
- **transport input ssh:** This move significantly improves the security posture of the switch. It instructs the switch to accept incoming connections only through SSH. As a result, it

indirectly turns off Telnet that is not encrypted, consequently securing the device against "eavesdropping."

Test Verification:



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l administrator 10.0.3.129

Password:

UNAUTHORIZED ACCESS PROHIBITED

System: Australia Branch Multi-layer Switch

This system is for the use of authorized users only.
Individuals using this computer system without authority,
or in excess of their authority, are subject to having
all of their activities on this system monitored and recorded.

IF YOU ARE NOT AUTHORIZED, DISCONNECT IMMEDIATELY.

AU_MLS>

[Connection to 10.0.3.129 closed by foreign host]
C:\>
```

As a part of the AU_MLS switch configuration verification, a remote access test was done through the PC AU-Mgmt-PC1 that is located in the Management VLAN 99 that is allocated to it. A secure session was established when the command below was given:

```
ssh -l administrator 10.0.3.129
```

This confirms the proper usage of RSA keys by the switch. The successful login and the displaying of the security banner reveal that the combination of the login local and transport input ssh commands has been correctly implemented to permit encrypted, credential-based access.

Once the user has been verified, he receives access to the User EXEC prompt (identified as AU_MLS>), where the execution of simple monitoring commands is mainly done. This step is a confirming sign that SSH is working at the maximum level, providing a protected and hidden space to start the administration tasks of the AU_MLS switch.

CHAPTER 11: RECOMMENDATIONS

The following section outlines three security recommendations to enhance the current network infrastructure of Alstom Networking Company:

11.1 Deployment of a Bastion Host at the India Remote Branch

It is recommended to deploy a bastion host at the India branch due to its role as a remote site that relies heavily on WAN connectivity for accessing centralized resources. A bastion host acts as a hardened gateway positioned in a demilitarized zone (DMZ), allowing controlled and monitored access to internal systems. By restricting exposed services and enforcing strict authentication policies, the bastion host minimizes the attack surface and reduces the risk of unauthorized access originating from external networks.

11.2 Implementation of a Next-Generation Firewall (NGFW) at KL-HQ

The Kuala Lumpur headquarters functions as the central hub for inter-branch communication, making it a critical point for security enforcement. Deploying a Next-Generation Firewall (NGFW) at this location enables advanced threat protection through application-aware filtering, intrusion prevention, and malware detection. This centralized security control improves visibility over network traffic and ensures consistent enforcement of security policies across all connected branches.

11.3 Deployment of a Honey Pot in the Australia Branch

The Australia branch, which includes wireless network infrastructure, presents a broader attack surface compared to fully wired environments. Deploying a honey pot at this site allows the organization to proactively detect and analyze malicious activities by attracting potential attackers to a controlled, non-production system. Monitoring interactions with the honey pot provides valuable insights into attack techniques and emerging threats, supporting improved incident response and overall network defense strategies.

CONCLUSION

The network design and execution as found in this report show a well-structured and thorough approach to the operational and security needs of a multi-site enterprise environment. The proposed architecture can successfully combine switching, routing, wireless networking, and Layer 2 security solutions to facilitate the geographically dispersed operations of Alstom by translating the given scenario into a workable network solution.

The IP addressing plan and logical topology were well-designed with the help of the VLSM in order to provide good utilization of addresses as well as the clear division between the departments and sites. Base networking facilities, including DHCPv4, dynamic and static routing, and centralized server implementation, make the process of communication among all the locations easy. EtherChannel and HSRP, which are high-availability methods, also increase the resilience of a network by minimizing single points of failure and offering service across the board.

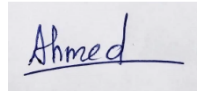
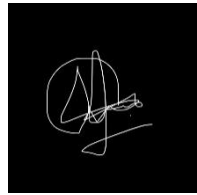
The wireless connection was introduced where necessary, which was through a WLC-based architecture with secure authentication, with a balance between ease of administration and high access control. Simultaneously, the Layer 2 security threats were addressed by implementing the right security measures, which strengthened the network against the typical attacks by LAN.

Altogether, the results of this assignment shed light on the essential role of conscious planning of a network, reasonable design choices, and safe configuration. The end product design is not only capable of fulfilling the technical needs of the situation, but it is also an expression of the principles of the real-world enterprise networking in accordance with the industry best practices.

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<https://www.techtarget.com/searchnetworking/definition/spanning-tree-protocol>

WORKLOAD MATRIX

Name (TP)	Tasks	Signatures
AHMED MIRAHUSAIN ALVI (TP084807)	• OSPF Router Configuration • Network Topology Design • IP-Addressing Table Design	
MOHAMED ABDIFATAH ALI (TP082459)	• India Branch • Server Farm Configuration • India DHCP Configuration • India Router-on-a-Stick Configuration	
ALAA AHMED MOHAMED ABDELHAI (TP084446)	• KL Branch • EtherChannel Configuration • KL DHCP Configuration • KL Router-on-a-Stick Configuration	
SULTAN ABDULLA OMAR TAKRORI (TP085327)	• Singapore Branch • Singapore Router-on-a-Stick Configuration • Voice VLAN Configuration • IOT Server Configuration	
HAMZA EL BJEIRMI (TP082392)	• HSRP Configuration • Switch Security Configuration • End-Device Labelling	
MUHAMMAD TAIMOOR JAMAL (TP085061)	• Australia Branch • WLC Configuration • Radius Server Configuration	